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SPECIFIC
TECHNICAL

Gas metering systems - Hourly gas metering devices - Interchangeability of flow meters≤ 10m3/h with point-to-point communication - Part 12-1: General and use cases

UNI/TS 11291-12-1

JANUARY 2020

Gas measurement systems - Hourly based gas metering systems - Interchangeability of meters with Q_{max}≤10m3/h with point to point communication - Part 12-1: General and use cases

The technical specification completes the technical regulatory requirements, defined in the applicable parts of the UNI/TS 11291 technical specifications, in order to allow the interchangeability of natural gas measuring groups intended for the "mass-market" (<G10) that communicate in point-to-point mode.
The technical specification defines the application functions and communication interfaces that must be made available by measurement groups and network devices on the local port for installation and maintenance purposes.

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PREMISE

The technical specification is reviewed every 3 years.
Any comments on UNI/TS 11291-12-1 must reach UNI by January 2022.

This technical specification was developed under the jurisdiction of the body federated to UNI

CIG - Italian Gas Committee

The UNI Central Technical Commission gave its approval on September 25, 2019.

This technical specification was ratified by the President of UNI and became part of the national regulatory body on 9 January 2020.

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INTRODUCTION

On October 22, 2008, the Italian Regulatory Authority for Energy, Networks and the Environment (ARERA) published Resolution ARG/gas 155/08, "Directive for the commissioning of gas metering units, characterized by minimum functional requirements and with remote reading or remote management functions, for delivery points of natural gas distribution networks." This resolution was repealed by Resolution 631/2013/R/GAS of December 27, 2013, as it was incorporated into Resolutions 575/2012/R/gas and 573/2013/R/gas. Amendments to the text of this technical specification may be necessary as a result of measures adopted by ARERA following that resolution.

The UNI/TS 11291-12 technical specification is composed of the following four parts, the full application of which is necessary for interchangeability purposes:

- UNI/TS 11291-12-1 Gas measurement systems - Hourly gas measurement devices - Interchangeability of meters with flow rate≤10 m3/h with point-to-point communication - Part 12-1: General and use cases.
- UNI/TS 11291-12-2 Gas measurement systems - Hourly gas measurement devices – Interchangeability of meters with flow rate≤10 m3/h with point-to-point communication - Part 12-2: Data model;
- UNI/TS 11291-12-4 Gas measurement systems - Hourly gas measurement devices - Interchangeability of meters with flow rate≤10 m3/h with point-to-point communication - Part 12-4: PP4 communication profile;
- UNI/TS 11291-12-6 Gas measurement systems - Hourly gas measurement devices - Interchangeability of meters with flow rate≤10 m3/h with point-to-point communication - Part 12-6: Test specifications for conformity evaluation.

Part 12-1 includes the overall architecture definition and the subset of use cases specified for interchangeability.

Part 12-2 defines the subset of the data model specified for interchangeability with the exception of media management strategies and modes.

Part 12-4 defines the strategies and methods for managing the transmission medium, the state machines for managing communication.

Part 12-6 defines a common framework for conformity assessment.

As regards local communication aspects, UNI/TS 11291-11-3 applies.

1 PURPOSE AND SCOPE

This technical specification completes the technical regulatory requirements, defined in the applicable parts of the UNI/TS 11291 technical specifications, in order to allow the interchangeability of natural gas measuring groups intended for the “mass market” (<G10) as well as other equipment operating in point-to-point networks.

This technical specification applies to GdMs serving the Delivery Points of distribution systems with a maximum capacity of less than 10 m3/h under reference conditions (or GdM <G10), which communicate with point-to-point protocol and which support one of the following communication networks: GPRS/UMTS/LTE or NB-IoT.

This technical specification defines the application functions and communication interfaces that must be made available by measurement groups and network devices on the local port for installation and maintenance purposes.

Additionally, local maintenance devices (HHTs), although intended to be interchangeable, are not based on the same hardware and are often supplied independently by distribution companies. For this reason, this document specifies standard interfaces on the local port to support local interventions interchangeably. However, it should be noted that defining the characteristics of HHTs is outside the scope of this technical specification.

This technical specification is to be used in conjunction with the UNI/TS 11291 series.

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Note For reasons of interchangeability, this technical specification includes detailed interpretations of the standards and specific use cases, objects, and data structures.
These elements are mandatory for interchangeability purposes, however care has been taken not to prevent manufacturers from providing extensions and alternative mechanisms permitted by the protocols: these, however, are not part of this technical specification.

2 REGULATORY REFERENCES

This technical specification refers, through dated and undated references, to provisions contained in other publications. These normative references are cited at the appropriate points in the text and are listed below. For dated references, subsequent amendments or revisions made to those publications are valid only if introduced into this technical specification as an update or revision. For undated references, the latest edition of the referenced publication (including updates) applies.

UNI/TS 11291-1:2013 Gas measurement systems - Gas measurement devices on an hourly basis - Part 1: General characteristics of the remote management or remote reading system

UNI/TS 11291-6:2013 Gas measurement systems - Gas measurement devices on an hourly basis - Part 6: Requirements for measuring groups with ranges less than 10 m^3/h (GdM <G10)

UNI/TS 11291-7:2013 Gas measurement systems - Gas measurement devices on an hourly basis - Part 7: Remote management systems for gas meters

UNI/TS 11291-10:2013 Gas measurement systems - Gas measurement devices on an hourly basis - Part 10: Safety

UNI/TS 11291-11-3 Gas measurement systems - Hourly gas measurement devices - Part 11-3: Local interface communication profile

UNI/TS 11291-12-2 Gas Metering Systems - Hourly Gas Metering Devices - Interchangeability of Meters with Flow Rates≤10 m^3/h with point-to-point communication - Part 12-2: Data model

UNI/TS 11291-12-4 Gas Metering Systems - Hourly Gas Metering Devices - Interchangeability of Meters with Flow Rates≤10 m^3/h with point-to-point communication - Part 12-4: PP4 communication profile

3 TERMS AND DEFINITIONS, ABBREVIATIONS

3.1 Terms and definitions

For the purposes of this document, the terms and definitions in IEC 62056-5-3, IEC 62056-6-1 and IEC 62056-6-2 and the following terms and definitions apply.

3.1.1 **interchangeability**: Ability to replace one device with another without reducing the original functionality and without malfunctions or loss of efficiency of the overall system.

3.1.2 **interoperability**: Ability of a system to exchange data with other systems of different types and/or from different manufacturers

3.1.3 **valid message, valid command**: A frame protected by a security key known to the specific recipient, which is syntactically and semantically correct and which passes integrity and security checks, and is accepted as such.

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Examples:

(1) A message sent to the public client is not considered valid in the context defined here.

(2) A message sent multiple times may be valid the first time but becomes invalid later when the frame counter replicates.

(3) A command authenticated with the wrong key is invalid.

3.1.4 **push service**Communication strategy where the data server sends a predefined set of information to a receiver with a defined timing strategy over a specific transmission medium.

3.1.5 **useful life**: Acceptable period of use in service. For the purposes of this specification, this period is considered to be equal to the validity period of the metric stamps and the CE marking, i.e., fifteen years.

3.2 Abbreviations

For the purposes of this technical specification the following abbreviations apply.

prospectus 1 **Abbreviations**

Abbreviation	Meaning	Meaning (in English)
ARERA	Regulatory Authority for Energy, Networks and the Environment	
APDU	Application Protocol Dataset	Application Protocol Data Unit
SAC	Central access system as defined in UNI/ TS 11291-1:2013	
FC	Plot Counter	Frame Counter
GPRS		General Packet Radio Service
GSM		Global System for Mobile communications
HHT		Hand-held terminal
ID	Identifier	
MU	Measurement Group (GdM)	
UC	Use case	Use Case
HLS		High Level Security
XML		Extensible Markup Language
EOB	End of billing	End of billing
EKSF	Key transport file	Electronic key shipment file
CA	Certification Authority	Certificate Authority
KEM		Key encapsulating mechanism
RNG	Random Number Generator	Random number generator
PEC	Certified Electronic Mail	
KEYC	Working or operating key (see UNI/TS 11291-10:2013)	
KEYS	Service key (see UNI/TS 11291-10:2013)	
KEYT	Installation/maintenance key (see UNI/TS 11291-10:2013)	

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4 GENERAL INFORMATION

This technical specification contains a list of use cases, which define the minimum mandatory subset required for interchangeability. The following criteria have been applied.

The use cases fully satisfy the requirements of Resolution ARG/gas 155/08 of the Electricity and Gas Authority, as subsequently amended and finally implemented by Resolution 631/2013/R/GAS of 27 December 2013.

Use cases indicate (only) one way to satisfy the requirements, normally selected based on simplicity of implementation. Where different implementation methods are available, they are permitted, provided they do not conflict with the requirements of this technical specification; however, implementation of the use cases as described in this technical specification remains mandatory.

Where, in very complex cases, two scenarios are defined, one basic and one alternative, both must be implemented for the purposes of interchangeability.

Due to the interchangeability of communication ports at certain levels, the use cases described from the remote communication perspective could generally also be applicable to the local port. Where differences exist, these are discussed in the "alternative scenarios" section.

This technical specification does not define any commercial use cases, such as switching vendors or activating or terminating a contract. Commercial use cases can be constructed using a more or less complex set of technical use cases described in this technical specification. Appendix D (informative) shows examples of this definition process.

For the purposes of this technical specification, it is also important to define a common reference for evaluating service levels. Indeed, these service levels, along with available network resources, are influenced by the communication strategies implemented by the various network components. This technical specification defines these reference levels in Appendix A (informative).

It should be clarified, however, that the aforementioned service levels are not intended to determine the success rate and, consequently, do not imply that the entire population of measurement groups belonging to a given installation is capable of meeting these objectives. Whenever a response or activation interval for a service is declared, it is intended as a time limit, beyond which the service must be declared inoperative.

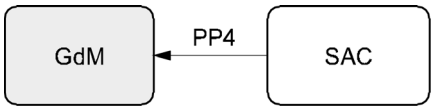
The above reference levels can also be used to define the reference conditions for defining the battery life of the various network components. For the purpose of defining battery life, additional reference conditions must be added. These additional parameters are defined in point A.1.

5 ARCHITECTURE

5.1 General information

This technical specification is based on the subset of the architecture defined in UNI/TS 11291-1:2013, and in detail as indicated in figure 1.

figure 1 **Domain of interchangeability**



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The GdM manages the relevant measurement functions as defined in UNI/TS 11291-6:2013 and makes the measurement and diagnostic functions available.

Furthermore, it can manage its own communication strategies (push) together with strategies that are managed by the SAC (pull).

In the following use cases it is therefore assumed that the SAC is responsible for managing the read strategies for:

- recovery of measurement data from the GdM;

- recovery of missing measurement data in any case (filling holes in the database);

- recovery of diagnostic data (event queues and/or special diagnostic logs) only for the GdMs involved, in case the condensed diagnostics identify suspicious or potentially faulty GdMs;

- manage the current actual state of the valve following a command.

5.2 Safety related elements

For security purposes, the encryption and authentication mechanisms provided by the communication protocol used are used.

5.3 Synchronization related items

The applicable parts of the UNI/TS 11291 series contain different requirements relating to synchronization, such as the maximum permissible drift, the maximum residual error after synchronization, and the time intervals used in the detection of timing errors, which can enable or prevent synchronization. For greater clarity, these requirements are summarized in Appendix B (normative).

6 USE CASES

6.1

General information

UNI/TS 11291-1:2013 defines the main use cases (under the definition of services) that fall within the scope and field of application of this technical specification.

In order to achieve interchangeability, a common understanding of use cases is required, and a single data model must underlie those use cases.

Furthermore, the use cases provided by UNI/TS 11291-1:2013, herein referred to as primary use cases, are very extensive, requiring a more detailed subdivision for clarity. In this technical specification, secondary use cases are derived from the aforementioned primary use cases.

In addition, the following basic services are identified:

- reading;

- writing;

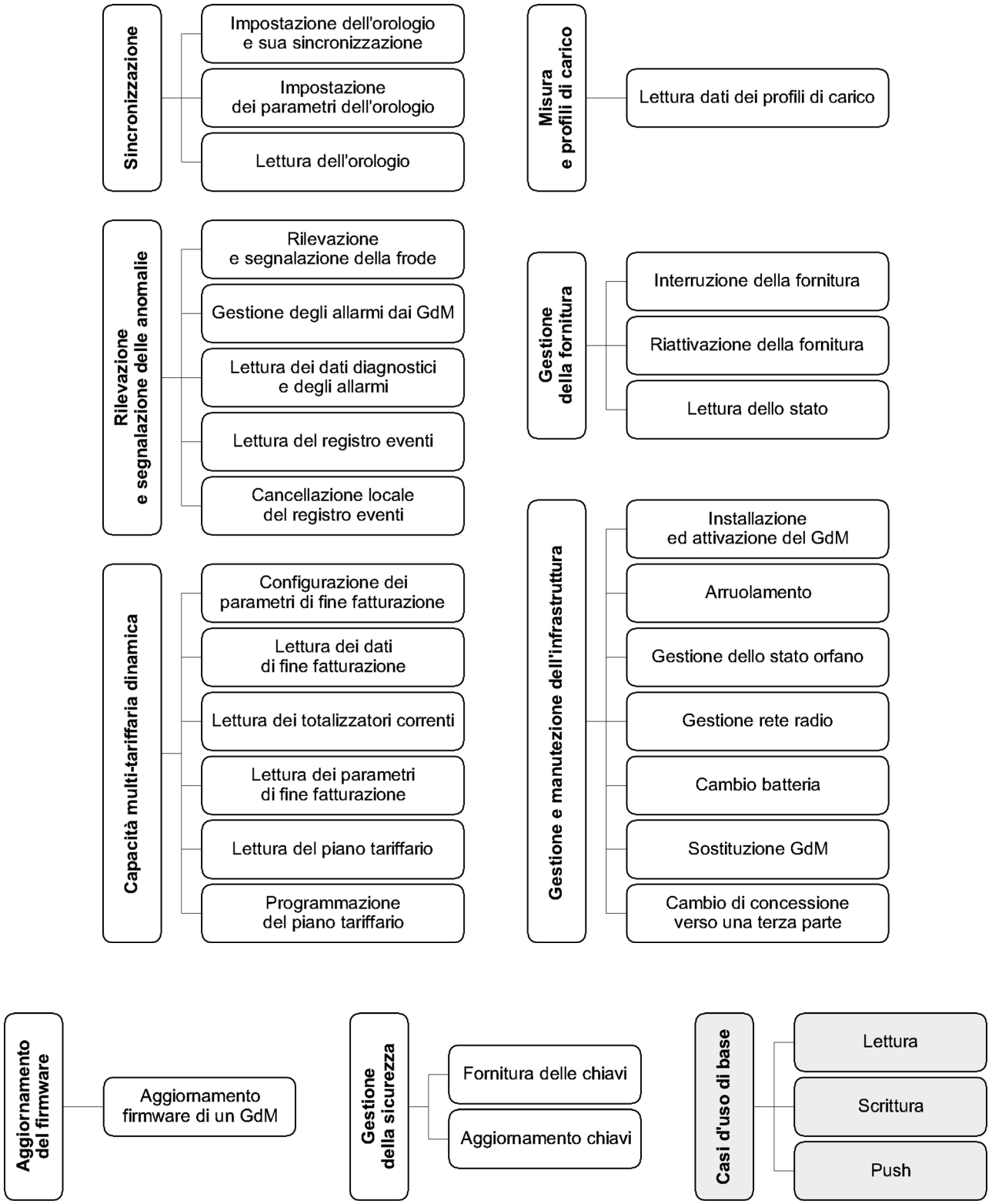
- push;

Some of the secondary use cases simply reduce to a basic service: they are therefore classified as basic use cases. They can all be described using a basic service, the only difference being the object and structure they bring with them.

Other use cases may require the execution of a complex sequence of actions, or require the definition of a complex data model, perhaps due to the need to optimize the amount of data exchanged. These are called complex use cases.

The following diagram in figure 2 shows the use cases, in hierarchical representation with respect to the services prescribed by UNI/TS 11291-1:2013.

figure 2 **Use case diagram**



6.2 Synchronization

6.2.1 General information

In the following sections the following hypotheses have been made:

a) daylight saving time (DST) is implemented at the first scheduled switch;

b) the storage records, the gas hour and the GdM display are expressed in local time.

6.2.2 Setting the clock and synchronizing it

The minimum set of commands for interchangeability must support both immediate synchronization and delayed synchronization across small clock drifts, when the synchronization information is sent via the time ("beacon") or a drift delta value.

prospectus 2 **Information Features**

Use case goals	The use case describes the setting and changing (i.e. synchronization) of the clock of a GdM (hereinafter referred to as "lower device") by another element (upper element).
Level	Primary
Prerequisites	- The lower device is equipped with a clock. - The lower device clock and time zone have already been set - The lower device has been installed and is operational. - The communication channel to the lower device (i.e. PP4) is available - Based on clock information from previous communications or on a periodic basis it has been decided that a synchronization command is necessary
Use case success condition	- The clock of the lower device is synchronized
Use Case Failure Condition	- The clock on the lower device is not synchronized
Primary roles	- SAC = upper element
Secondary roles	- Subordinate network device = inferior device - GdM
Triggering cause	The top element makes a request to the partnered bottom device to synchronize the clock
Hypothesis	- The use case does not change whether the synchronization is immediate or deferred (drift). - Communications are made secure. The GdM has started communication (push use cases (section 6.10.3) or is listening for a synchronization command. It is the SAC's responsibility to estimate the network propagation times that must be considered when calculating the difference between the GdM clock and the reference clock.
Security requirements	Prevent entities external to the system from being able to send a clock set command; the synchronization command is a critical command that requires at least authentication.
Category	Synchronization
References	- UNI/TS 11291-1:2013, point 4.2 "Services" - UNI/TS 11291-1:2013, point 4.3.1 "Service: Synchronization" - UNI/TS 11291-6:2013, point 7.5 "Clock", in particular 7.5.2 Synchronization" (GdM) - UNI/TS 11291-7:2013, point 4.3.2 "Clock synchronization" (General)

The following scenarios are

considered: a) Main success scenario

10) The top element queues the command *clock synchronization* for its affiliated lower devices.

20) When the communication channel to the lower device is available for the upper element (for example when the lower device makes a push or if there is a communication window programmed with the lower device) the upper element sends the command *clock synchronization* to the lower device.

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30) The lower device receives the command, checks the difference of its clock with the reference received in the command.

40) The lower device aligns its clock immediately or in steps according to the rules defined in the relevant parts of the UNI/TS 11291 series (within the accepted limits).

42) The SAC may decide to send a command to the lower device that performs a specific time shift; this shift is computed by the SAC as the difference between the date-time of receipt of a message and the date-time of the message entered by the sender.

50) The lower device handles all further services affected by the time change following the rules specified in Appendix B.

60) The lower device sends a confirmation to the upper element.

70) The top element receives confirmation.

80) The use case ends.

b) Alternative scenarios

The alternative scenario addresses local synchronization, which should be performed during normal operation. Local synchronization during installation is outside the scope of this use case and is subject to special behavior of the GdM, as described in the installation use case (see section 6.7.1).

- The operator locally connects a handheld to the device, having the appropriate rights to do so
- The operator sends a command via a handheld device *clock synchronization* to the lower device
- The lower device receives the command, checks the difference of its clock with the reference received in the command
- The lower device aligns its clock immediately or in steps according to the rules defined in the relevant parts of UNI/TS 11291 (within the accepted limits).
- The lower device handles all further services affected by the time change following the rules specified in Appendix B.
- The operator verifies the proper reception and execution of the command
- The use case ends.

prospectus 3 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements. The misalignment between the upper element and the lower device must not be greater, after synchronization, than the limits imposed by UNI/TS 11291-1:2013
Frequency	according to the prescribed SLA.
Subordinating use case	Nobody
Subordinate use cases	- Use case point 6.10.1: Reading - Use case point 6.10.2: Writing - Use case point 6.10.3: Push

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6.2.3 Setting the clock parameters

prospectus 4 **Features information**

Use case goals	The use case describes setting parameters related to clock behavior (including daylight saving time handling)
Level	Primary
Prerequisites	- The lower device is equipped with a clock. - The lower device has been installed and is operational. - The communication channel is available
Use case success condition	- Timestamps are written in official local time (including daylight saving time information)
Use Case Failure Condition	- Timestamps are not written in official local time (including daylight saving time information)
Primary roles	- SAC (remote)
Secondary roles	- GdM
Triggering cause	The SAC sends a request to the lower device to set the device's clock to local time and to program the daylight saving time parameters.
Hypothesis	- Local time is activated - Local time conversion is enabled (e.g. from the time zone DB) - The device is operating in local time and daylight saving time management is enabled
Security requirements	The clock setting must be protected from external actions (authenticated)
Category	Synchronization
References	- UNI/TS 11291-1:2013, point 3 - UNI/TS 11291-1:2013, point 4.3.5 - UNI/TS 11291-6:2013, point 7.5.1 - UNI/TS 11291-6:2013, point 7.3

The following scenarios are

considered: a) Main success scenario

10) The use case starts when the SAC queues a request for the lower device to set parameters related to clock management (e.g. daylight saving time)

20) When the communication channel with the GdM is available the SAC transmits the command to the GdM

30) The GdM receives and validates the command

40) The GdM confirms to the SAC that the command has been received and executed successfully. The time (Universal Time) is converted to local time (including daylight saving time).

50) The use case ends.

b) Alternative scenarios:

- The PP4 communication channel is not available (step 20 fails).
- Command validation by the GdM fails (step 30 failure).
- The GdM is unable to execute the command (step 30 fails).
- The SAC does not receive confirmation (step 40 failure).

prospectus 5 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	At the first programming
Subordinating use case	Nobody
Subordinate use cases	- Use case point 6.10.1: Reading - Use case point 6.10.2: Writing - Use case point 6.10.3: Push

6.2.4 Reading the clock

Explicit reading of the clock is not expected because the clock information must already be present in information exchanges related to many other use cases.

6.3 Detection and reporting of anomalies

6.3.1 Fraud detection and reporting

The technical specifications of the UNI/TS 11291 series do not prescribe any real-time alarms, i.e., immediate alarm pushes, so there is no difference between diagnostics and alarms except for the position of the relevant bits in the diagnostic register. For this reason, and to ensure SLAs, the diagnostic status must be sent in all possible transmissions.

Furthermore, for greater service availability, the GdM provides dynamic connection strategies depending on the system status, with the criterion of reducing the frequency of communications when a less stringent SLA is applicable.

Regarding communication strategies, further details are provided in point A.2.

A specific “diagnostic object” must be defined in the data model, bit by bit within the scope of interchangeability, defining the behavior of the GdM in setting/resetting the related bits.

Additionally, there are some special counter logs that add up significant events or time spent on critical activities (for example, the number of incorrect logins or total synchronized seconds). These logs would require significant network resources if they had to be transmitted every time, but they can provide significant information, especially regarding fraud attempts. Therefore, these logs must be periodically read by the SAC.

In the field of interchangeability, the following registers are standardized, with reference to the different parts of the UNI/TS 11291 series.

- Number of seconds elapsed since the GdM was first switched on (point 6.4.2 of UNI/TS 11291-10:2013).

- Number of accesses with incorrect credentials (point 10.6.5.2 of UNI/TS 11291-10:2013).

- R1 = GdM of synchronizations (count of the number of synchronizations performed)

- R2 = GdM of seconds added during synchronizations (number of seconds shifted in the clock direction)

- R3 = GdM of seconds subtracted during synchronizations [number of seconds shifted in the opposite direction of the clock (for R1,R2,R3 see point 7.5.2 of UNI/TS 11291-6:2013)]

- Remaining battery life (see point 9 of UNI/TS 11291-6). Many events are generated by

the GdM itself or determined by the environment in which it operates.

Some events determined by the GdM's operating environment, and not generated by interactions with the SAC or the local operator (for example, those associated with fraud attempts), may occur with high frequency. To avoid saturating the relevant log, these events must be recorded only upon their first occurrence during the gas day. The detailed list of events to be managed in this manner is contained in the data model specification.

Each event has a Unique Event Identifier that allows it to be uniquely identified. Each event is assigned to one of the two event logs and is stored only in that one. This assignment is fixed and cannot be changed dynamically. Capturing these events in application objects may require obtaining, in addition to the related timestamp, related values, where applicable, such as the current totalizer.

Events can also set and clear relevant bits in the diagnostic register.

Reading the event log must neither clear the contents of the event log nor change the state of any status bits related to the related event.

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6.3.2 Alarm management from GdMs

This is a complex use case that defines how the system behaves when it sees an alarm in the diagnostic object.

Within the scope of this standard, the mode that provides for the generated alarm to be acquired by the SAC by reading the diagnostics during a scheduled connection is active.

6.3.2.1 Alarm management from GdMs

prospectus 6 **Features information**

Use case goals	The use case describes reading alarm management from the GdM
Level	Primary
Prerequisites	The GdM was enrolled in the SAC.
Use case success condition	The alarm information is received by the SAC.
Use Case Failure Condition	The alarm information is not received by the SAC.
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The alarm condition occurs as per point 7.6 of UNI/TS 11291-6:2013.
Hypothesis	The GdM has sufficient coverage to support reliable communications on the PP4 interface. NOTE: An explicit command to reset the alarm bit(s) is not mandatory. Furthermore, in the case of alarms where the alarm status is cleared when the alarm condition is no longer present, the SAC would not receive notification of the event. For this reason, a status register must exist that also provides the diagnostic history where the alarm bits are not reset when the triggering event disappears, and which is reset at the end of each period.
Security requirements	High, protected from external actions (authenticated)
Category	Alarm detection and management.
References	- UNI/TS 11291-1:2013, point 4.3.9. - UNI/TS 11291-6:2013, points 7.1 and 7.6.

The following scenarios are

considered: a) Main success scenario

10) The use case starts with an alarm condition that occurred in the GdM. See point 7.6 of UNI/TS 11291-6:2013 for possible conditions.

20) The GdM records the alarm in the diagnostic register and in the event log. See point 7.1 of UNI/TS 11291-6:2013. This means that at least one of the bits defined in the diagnostic register becomes active.

30) At the first expected communication window, the GdM transmits, in one or more of the messages exchanged with the SAC, the diagnostic register which indicates that an alarm event is active.

40) The SAC detects an alarm condition following examination of the diagnostic log.

50) Optionally, the SAC acquires the event log in its entirety, or in part through selective access, from the GdM according to use case point 6.3.3 (Reading the GdM event log)

60) The use case ends.

prospectus 7 **Additional Information**

Priority	See Appendix A.
Expected performance	UNI/TS 11291-1:2013 requires that alarms be made known within 24 hours
Frequency	Depending on the frequency of message exchange.
Subordinating use case	Nobody.
Subordinate use cases	Use Case Point 6.11: Basic Services

6.3.3 Reading diagnostic data and alarms

The default behavior of the GdM is to insert a diagnostic log into messages to the SAC. This allows the system to perform basic actions based on it. Additionally, this use case handles situations where the SAC needs to retrieve extensive diagnostic information for analysis purposes.

prospectus 8 **Features information**

Use case goals	This use case is responsible for reading the extended diagnostic data and alarm logs of the GdM
Level	Primary
Prerequisites	The GdM was enrolled in the SAC.
Use case success condition	Diagnostic and alarm registers are read from the SAC
Use Case Failure Condition	Diagnostic and alarm logs are not read by the SAC
Primary roles	SAC
Secondary roles	GdM
Triggering cause	On request: The SAC requests alarms and/or diagnostic data from a GdM (see UC - reading).
Hypothesis	The GdM has sufficient coverage to support reliable communications on the PP4 interface. NOTE: The bits in the diagnostic register are recorded according to an agreed-upon format.
Security requirements	High
Category	Anomaly detection and reporting.
References	UNI/TS 11291-1:2013, point 4.3.9 UNI/TS 11291-1:2013, point 7.6

The following scenarios are

considered: a) Main success scenario

10) This use case begins when the SAC issues a secure command to retrieve alarm log data and other diagnostic information from the GdM.

20) When the communication channel with the GdM is available, the SAC transmits the command to the GdM.

30) The GdM receives the command and sends the contents of the requested registers to the SAC.

40) The SAC receives the alarm or diagnostic information and processes it.

50) The use case ends.

prospectus 9 **Additional Information**

Priority	See Appendix A.
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use cases	Nobody
Subordinate use cases	UC point 6.10.1: reading

6.3.4 Reading the event log

Introduction: The use case requires a standard definition of the event log format as a precondition. For the purpose of interchangeability, two event logs are defined: a metrological one, which records only the relevant events defined by UNI/TS 11291-6:2013 and UNI/TS 11291-10:2013, and a non-metrological one. This use case refers to the metrological one.

Messages sent by the GdM generally also include the number of recorded events; this allows the SAC to recognize if new events have been recorded and proceed with their reading.

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prospectus 10 **Features information**

Use case goals	This use case is responsible for reading events present in the GdM event log.
Level	Primary
Prerequisites	The GdM has sufficient coverage to support reliable communications on the PP4 interface. The GdM has already been enlisted by the SAC.
Use case success condition	Event log events are available in the SAC
Use Case Failure Condition	Event log events are not available in the SAC
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC has detected, using the number of recorded events, that there are new events in the GdM since the last communication.
Hypothesis	Messages sent by the GdM on a periodic basis always report the number of events stored in the event log.
Security requirements	High
Category	Detection and reporting of anomalies
References	UNI/TS 11291-1:2013, point 4.3.9 UNI/TS 11291-6:2013, point 7.1

The following scenarios are considered: a) Main success scenario

10) The use case begins when the SAC generates a request to read a GdM's event log. The SAC has detected that new events have been logged by the GdM by reading the related GdM previously sent by the GdM.

20) The SAC issues a read command to retrieve the section of the event log containing the new events.

30) When the communication channel with the GdM is available, the SAC transmits the command to the GdM.

40) The GdM receives the command and sends the requested section of the event log to the SAC.

50) The SAC receives the information and processes it.

60) The use case ends.

prospectus 11 **Additional Information**

Priority	See Appendix A.
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	Nobody
Subordinate use cases	UC point 6.11.1: reading

6.3.5 Clearing the event log

prospectus 12 **Features information**

Use case goals	This use case is responsible for clearing the GdM event log.
Level	Primary
Prerequisites	The SAC, or a previously authorized operator, has the credentials to perform the deletion operation
Use case success condition	The event log is completely cleared. Only one record is left: the one that records the event that cleared the log. All events prior to the clearing are returned.
Use Case Failure Condition	The event log is not cleared
Primary roles	The operator with HHT (connected to the GdM through the local port), SAC
Secondary roles	GdM
Triggering cause	The delete command is sent by HHT or SAC.
Hypothesis	The operation is managed safely in accordance with the provisions of UNI/TS 11291-10:2013 The use case described here involves the use of the maintenance key. The same goal can also be achieved using the operation key, without needing to enable the maintenance session. Furthermore, only the case where the maintenance key deactivates due to timeout is considered.
Security requirements	High
Category	Anomaly detection and reporting
References	UNI/TS 11291-1:2013, point 4.3.9 UNI/TS 11291-6:2013, point 7.6

The following scenarios are

considered: a) Main success scenario

10) The use case starts when the SAC enables credentials for the HHT to activate a maintenance session and provides access rights to perform the delete operation.

20) HHT activates the maintenance session and reads the contents of the event log.

30) HHT generates a reset command for the event log.

40) The GdM receives the command and deletes the entire event log. It automatically generates the "event log deleted" event.

50) The GdM returns confirmation to the HHT that the command has been correctly executed

60) The number of events recorded in the event log is set to “1”

70) The use case ends.

b) Alternative scenario

- In the GdM, the HHT credentials have not been enabled or the access rights for this operation are not enabled.

This alternative scenario is responsible for handling the situation when the SAC is unable to ensure the activation of the maintenance session and/or grant access rights for this operation, for example by not issuing the correct maintenance key.

- The HHT sends the reset command for the event log or attempts to activate a maintenance session but the GdM returns a refusal to the HHT.
- The event log reset function is not possible.

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prospectus 13 **Additional Information**

Priority	See Appendix A.
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	Nobody
Subordinate use cases	-

6.4 Dynamic multi-tariff capacity

For the purposes of interchangeability, the minimum number of EOB records that must be recorded and read must be defined.

Considering the different scenarios in which EOBs can be recorded asynchronously, the minimum number must be at least 20, to account for 13 monthly invoicings and 7 asynchronous closings.

6.4.1 Configuring End of Billing (EOB) parameters

This UC also includes the “trigger” configuration for immediate EOB.

The UC includes the EOB configuration: period divisor and start date (if periodic), date and time (if asynchronous), and EOB reason. The UC must also include the configuration of the conventional end-of-day time, as this is related to it.

prospectus 14 **Features information**

Use case goals	This use case is responsible for configuring an EOB interval according to multi-tariff registration
Level	Primary
Prerequisites	Nobody
Use case success condition	EOB has been configured, ToU is consistent
Use Case Failure Condition	EOB cannot be configured
Primary roles	GdM SAC
Secondary roles	
Triggering cause	The request to modify the EOB of a GdM was sent by the SAC
Hypothesis	ToU data is stored in two distinct lists such as the following: Period | Date | ToU1 | ToU2 | … | ToUn Current period (Period P0) | D(0) | V(0,1) | V(0.2) | … | V(0.3) Updated every day or upon request Period | Date | ToU1 | ToU2 | … | ToUn Previous Period (Period P1) | D(1) | V(1,1) | V(1,2) | … | V(1,3) … | … | … | … | … | … PM | D(m) | V(m,1) | V(m,2) | … | V(m,n) The previous billing period was managed in FiFo mode Note The above structure is just an example. The database definition is found in UNI/TS 11291-12-2.
Security requirements	Nobody
Category	Multi-tariff
References	-

The following scenarios are

considered: a) Main success scenario

10) The SAC queues a command to configure a new EOB for a GdM it manages (see also UC point 6.11.2 Writing)

20) When the communication channel with the GdM is available, the SAC transmits the command to the GdM.

30) The GdM receives the command

40) The date for the activation of the new EOB is stored for the next execution

50) An event is generated to record the setting change

60) The GdM confirms to the SAC that the command has been correctly received

70) The SAC receives confirmation

80) The use case ends.

b) Alternative scenarios

- There is no confirmation from the GdM.

- The command cannot be accepted (for example, because the event log is full).

prospectus 15 **Additional Information**

Priority	See Appendix A.
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	Nobody
Subordinate use cases	UC point 6.10.2 Writing

6.4.2 Reading the end-of-billing data

The structure of the end-of-billing data archive is defined in UNI/TS 11291-12-2.

prospectus 16 **Features information**

Use case goals	This use case is responsible for reading the end-of-billing data
Level	Primary
Prerequisites	None
Use case success condition	The data is read
Use Case Failure Condition	The data is not read
Primary roles	SAC
Secondary roles	GdM
Triggering cause	One or more new end-of-billing records have been detected (for example, by detecting an increment in the billing period counter). Or, a push strategy has been programmed that sends them without an explicit request from the SAC.
Hypothesis	The information of the end of billing period is stored in a list managed in FiFo mode NOTE: The definition of the structure is found in UNI/TS 11291-12-2.
Security requirements	Nobody
Category	Multi-tariff
References	

The following scenarios are

considered: a) Main success scenario

10) The SAC prepares a read command if necessary to recover the missing end-of-billing data (according to UC point 6.11.1 Reading).

20) At the scheduled time the GdM opens the communication channel with the SAC.

30) The GdM sends, if programmed, the data contained in the end-of-billing archives.

40) The SAC transmits the read command to the GdM if necessary.

50) The GdM responds with the requested data.

60) The SAC receives the information and processes it.

70) The use case ends.

b) Alternative scenario

- No cases other than those already covered by UC point 6.10.1 Reading.

prospectus 17 **Additional Information**

Priority	See Appendix A.
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	-
Subordinate use cases	UC point 6.10.1 Reading

6.4.3 Reading the current totalizers

This use case uses basic reading.

For values that become valid at the end of a period, there is generally a temporary register used to calculate these values, such as the maximum conventional converted flow rate. These registers must be updated at specific time periods for interchangeability purposes. These periods are defined as follows for the main variables:

- The maximum converted conventional flow rate register must be updated every 5 minutes.

- For all other similar variables, the requirement of UNI/TS 11291-6:2013 applies (i.e. every day).

- Any period diagnostic must contain all diagnostic bits that occurred from the end of the previous period up to the time of reading.

6.4.4 Reading End of Billing (EOB) parameters

prospectus 18 **Features information**

Use case goals	This use case is responsible for reading billing period parameters
Level	Primary
Prerequisites	Nobody
Use case success condition	The parameters are read
Use Case Failure Condition	The parameters are not read
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC needs to know the configuration of the billing period parameters, for example for a periodic verification of the configuration, following maintenance, after enrollment of a new meter, upon request for verification by the end customer (see UC use case point 6.11.1 basic reading)
Hypothesis	NOTE: The definition of the parameters can be found in UNI/TS 11291-12-2
Security requirements	Nobody
Category	Multi-tariff
References	-

The following scenarios are

considered: a) Main success scenario

10) The SAC provides a read command for the end-of-billing parameters (according to UC point 6.11.1 Reading).

20) When the communication channel with the GdM is available, the SAC transmits the command to the GdM.

30) The GdM responds to the command with the end-of-billing parameters.

40) The SAC receives the information and processes it.

50) The use case ends.

b) Alternative scenario

- No cases other than those already covered by UC point 6.10.1 “Basic reading”

prospectus 19 **Additional Information**

Priority	See Appendix A.
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	
Subordinate use cases	UC point 6.10.1 Reading

6.4.5 Reading the tariff plan

prospectus 20 **Features information**

Use case goals	This use case is responsible for reading the current rate plan
Level	Primary
Prerequisites	None
Use case success condition	The current tariff plan is read
Use Case Failure Condition	The current tariff plan is not read
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC needs to know the tariff plan configuration, for example for a periodic configuration check, following maintenance, after enrollment of a new meter, upon request for verification by the end customer (see UC use case point 6.11.1 basic reading)
Hypothesis	The definition of the parameters can be found in UNI/TS 11291-12-2.
Security requirements	Nobody
Category	Multi-tariff
References	-

The following scenarios are

considered: a) Main success scenario

10) The SAC has issued a read command to the GdM (according to UC point 6.11.1 Basic reading)

20) The GdM forwards the response to the SAC

30) The use case ends

b) Alternative scenario

- No cases other than those already covered by UC point 6.10.1 Reading

prospectus 21 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	-
Subordinate use cases	UC point 6.10.1 Reading

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6.4.6 Tariff plan programming

prospectus 22 **Features information**

Use case goals	This use case is responsible for scheduling the current rate plan
Level	Primary
Prerequisites	None
Use case success condition	The new tariff plan is activated
Use Case Failure Condition	The new tariff plan is not activated
Primary roles	SAC
Secondary roles	GdM
Triggering cause	On request (see UC use case point 6.11.2 basic writing)
Hypothesis	The ToU data is stored in two distinct lists such as the following Period | Date | ToU1 | ToU2 | … | ToUn Current Period (Period P0) | D(0) | V(0,1) | V(0.2) | … | V(0.3) Updated daily or upon request Period | Date | ToU1 | ToU2 | … | ToUn Previous Period (Period P1) | D(1) | V(1,1) | V(1,2) | … | V(1,3) … | … | … | … | … | … PM | D(m) | V(m,1) | V(m,2) | … | V(m,n) The tariff plan defines which ToU should be used for any given time of day (it may also take into account Saturdays, Sundays, and national holidays). NOTE The above structure is just an example. The definition of this and other databases can be found in UNI/TS 11291-11-2.
Security requirements	Nobody
Category	Multi-tariff
References	-

The following scenarios are

considered: a) Main success scenario

10) The SAC has sent the command to set a new tariff plan to the GdM (according to UC point 6.11.2 basic writing)

20) The GdM receives the command

30) The GdM overwrites the “passive tariff plan”, i.e. it configures the future tariff plan

40) The event is recorded in the event log to indicate the storage of the passive tariff plan

50) The GdM sends the information to the SAC that the command has been executed correctly

60) The rate plan is always activated at the gas end-of-day time unless the GdM is in the "not configured" state and the activation date is in the past, in which case activation is immediate. Based on the scheduled date, activation occurs on the first occurrence of the gas end-of-day time (date in the past) or at the end of the scheduled day (date in the future).

70) When the tariff plan is activated, the related event is stored (as per the requirement of UNI/TS 11291-6:2013, according to the codes reported in UNI/TS 11291-12-2).

80) The use case ends.

b) Alternative scenario

The GdM does not report a positive outcome at step 50. The other steps from 30 to 70 are therefore not performed.

prospectus 23 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	-
Subordinating use case	-
Subordinate use cases	UC point 6.10.2 Writing

6.5 Measurement and load profiles

In the scope of this technical specification, “load profile” means a vector of elements consisting of totalizers, accompanied by the relevant period diagnostics and time stamps, recorded on a daily basis.

6.5.1 Reading load profile data

Load profiles are available for reading by the SAC and can be retrieved using the basic "profile read" service. This service should only be used to recover data missing from the profile reconstructed in the SAC, and not already retrieved using other services.

Note The concepts of “compact array” and “compact weave” can be usefully applied to carry load profiles.

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6.6 Supply Management

6.6.1 Supply interruption

prospectus 24 **Features information**

Use case goals	Supply interruption
Level	Primary
Prerequisites	The GdM has been installed and is fully operational at the customer's premises. Access via PP4 is available. The SAC knows the credentials used for the current security of the GdM. The valve's operating modes have been programmed (e.g., behavior in the event of fraud, use or not of the PIN when reopening, behavior on holidays, etc.)
Use case success condition	The gas flow is interrupted
Use Case Failure Condition	The gas flow is not interrupted
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC receives a request to close the valve from an authorized entity. For example, the end customer terminates the contract because they are leaving the home, or due to non-payment, or because the gas supply is seasonal.
Hypothesis	For various reasons, but with the express exclusion of safety functions, it must be possible to send a command to close the integrated shut-off valve. It is assumed that a secure connection is established between the SAC and the GdM. The SAC has preliminarily programmed the communication strategies filter mask and the S00 and S01 communication strategies in the GdM. The GdM uses the S00 communication strategy (i.e. the valve is open). The command to be sent to the GdM must contain at least: - Execution date - Valve reopening PIN - Duration of validity of the opening command - Possible asynchronous closure of the billing period
Security requirements	Command always encrypted and authenticated
Category	Supply Management
References	-

The following scenarios are considered: a) Main success scenario

10) The distributor receives a request from the seller to close the valve.

20) The distributor programs a valve closing order in the SAC; The command to be sent to the GdM must contain at least:

- Execution date
- Valve reopening PIN
- Duration of validity of the opening command
- Possible asynchronous closure of the billing period

30) The GdM connects to the SAC according to the current strategy.

40) During the connection with the GdM, the SAC sends the closing command to the GdM.

Note 1 By appropriately reprogramming the communication strategy, it is possible to revoke the command before it is implemented.

50) The GdM receives the command and confirms successful reception to the SAC. The GdM schedules the valve to close according to the delay set by the SAC.

60) At the programmed instant the GdM closes the valve, modifies the corresponding status bits and activates the S01 connection strategy.

Note 2 The strategy should contain at least the following information:

- valve status;
- reason for closure (e.g. by command);
- any anomaly found.

If the scheduled time falls on a NON-useful day and the related function is not expressly disabled, the closing action must be postponed to the beginning of the first subsequent useful day.

Note 3 The “useful” day: is defined as each day of the week other than the days indicated as holidays, including the patronal feast day of the municipality in which the return point is located, Saturdays and the days preceding Saturdays or holidays.

70) The GdM, during the period in which the valve is closed, checks for gas flow. If a flow is detected, it activates the appropriate diagnostics.

Note 4 The minimum detectable flow depends on the construction characteristics of the GdM.

80) The use case ends

b) Alternative scenario

The command is forwarded via local communication.

prospectus 25 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	-
Subordinate use cases	-

6.6.2 Reactivation of supply

prospectus 26 **Features information**

Use case goals	Reactivation of supply
Level	Primary
Prerequisites	The GdM has been installed and is fully operational at the customer's premises. Access via PP4 is available. The SAC knows the GdM's latest operating key. The valve operating modes have been programmed (e.g. use or not of the PIN when reopening, etc.)
Use case success condition	Gas flow is restored
Use Case Failure Condition	Gas flow is not restored
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC receives a request to enable the valve reopening.
Hypothesis	The SAC operator requests authorization to restore the gas supply, within a predefined time window, in order to allow an operator or the end customer to restore the supply locally. The SAC has preliminarily programmed the communication strategies filter mask and the communication strategies S00, S01 and S02 in the GdM. This enabling does not equate to re-establishing gas flow: a subsequent local confirmation action is required for this purpose, provided that this is not prevented by other reasons.
Security requirements	Command always encrypted and authenticated
Category	Supply Management
References	-

The following scenarios are

considered: a) Main success scenario

10) The seller sends the distributor a supply restoration order.

20) The Distributor programs in the SAC an order to enable the opening of the valve with a specific execution date and a validity period starting from the execution date.

30) The GdM connects to the SAC according to the current strategy, i.e. S01.

40) During the connection with the GdM, the SAC sends the GdM the command to enable opening. It can then close the connection; the communication strategy remains S01.

50) Once the command has been received, at the programmed time and for the programmed duration, the GdM signals (on the display) that the valve is enabled for reopening.

60) The GdM awaits the local command, implemented by action by an operator or the end user (for example pressing a button).

70A) When the customer performs the requested action on the GdM, such as pressing a button, the valve is opened.

80A) If the opening conditions are met (e.g. by means of the opening leak test) the valve remains open.

90A) The status changes to “valve open”

100A) together with the time stamp, the related event is recorded in memory. The

110A) GdM activates the S00 communication strategy.

120A) The use case ends

70B) If the time window expires without the local opening command being given, the GdM returns to the “valve closed” state and the communication strategy remains S01.

80B) The valve state remains closed and at least one “end of reconnection window” event is recorded along with its timestamp.

90B) The use case ends.

80C) If the valve opening leak test fails, action 70A is repeated two more times.

90C) After three failed opening attempts due to leaks, the state changes to “valve closed” and the communication strategy remains S01.

100C) At least one “valve closed due to leak” event along with its timestamp is recorded in memory.

110C) The use case ends.

b) Alternative scenario 1

The command is forwarded via local communication

prospectus 27 **Features information**

Triggering cause	The operator is sent to the site and performs the restoration of the gas supply using a safe procedure.
Hypothesis	The GdM has been installed and is fully operational at the customer's premises. Access to the installation site is guaranteed. The valve is closed and the GdM is working properly.
Security requirements	The operator must be authenticated

10) The operator, in order to operate on the valve, authenticates himself (via HHT and local interface)

20) The operator sends a command, via the local interface, to enable valve reopening with a specific execution date and time and a valid duration. If immediate activation is desired, the execution date and time must be set in the past.

30) The GdM sends information to HHT that the command has been received and is valid.

The use case evolves as in the main scenario starting from point 50 and with all the foreseen variants.

c) Alternative scenario 2

The enabling command is sent by entering a PIN code.

The GdM must be enabled to reopen the valve using a password; the indication that the supply can be restored must be shown on the display but only during the time available for reopening.

prospectus 28 **Features information**

Triggering cause	The person operating on-site is on site and has been made aware of the value of the PIN.
Hypothesis	The GdM has been installed and is fully operational at the customer's premises. Access to the installation site is guaranteed. The valve is closed and the GdM is working properly. Remote communication (via PP4) is not available, i.e. the GdM is adopting the S02 communication strategy. The PIN and the validity window of the opening commands have been previously programmed in the GdM. NOTE: The PIN reopening enablement can also be used in the presence of communication on PP4, and in this case the main scenario occurs with the only difference that the PIN, if entered correctly, replaces the reopening enablement command from SAC via PP4 or local port.
Security requirements	-

10) The seller sends the distributor a supply restoration order.

20) The value of the PIN is communicated to the person on site.

30A) The person on site enters the PIN**correctly**.

40A) The GdM enables the valve to open and activates the relevant signal on the display.

60A) The GdM waits for the opening command, implemented by action by the person on site (for example pressing a button).

70A) When the person on site performs the required action on the GdM, such as pressing a button, the valve is opened.

80A) If the opening conditions are met (e.g. by means of the opening leak test) the valve remains open

90A) The status changes to “valve open”

100A) Together with the time stamp, the related event is recorded in memory.

110A) The GdM remains in the S02 connection strategy.

120A) The use case ends

30B) The person on site enters the PIN**not correctly**for the maximum number of scheduled attempts.

40B) The GdM invalidates the PIN and the valve remains closed.

50B) The use case ends.

prospectus 29 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements
Frequency	If necessary
Subordinating use case	-
Subordinate use cases	-

6.6.3 Reading the status

Valve status information evolves much more slowly than the exchange of information and commands between the GdM and the SAC. It is therefore advisable for the SAC to refresh the information it contains as frequently as possible. To this end, the data model ensures that all the most frequently performed information exchanges contain valve status information.

6.7 Infrastructure management and maintenance

6.7.1 Installation and activation of the GdM

One requirement on which the use cases are based is that, when the installer leaves the field, the GdM is able to measure and manage the information necessary for billing.

Since it is not possible to guarantee during installation that the network coverage is sufficient and/or the GdM is immediately available for enlistment by the SAC, it may be necessary to use a local terminal (HHT) to program parameters in the GdM (for example PDR, date and time) and switch the GdM from the non-configured state to the normal state.

To streamline installation procedures and allow for the use of less qualified personnel, an installation method is defined that does not require the use of the terminal. Please note that, in this case, switching to normal mode and programming the PDR are not guaranteed in the absence of network coverage.

The following use case describes a process that includes a network coverage check.

It should be noted that this process may require personnel to remain in the field for longer.

A shorter alternative process is to leave the GdM in the normal periodic auto-connect mode and let the enrollment evolve according to the relevant use case (Enrollment point 6.7.2).

The following use case considers both possibilities. The short process requires omitting steps 50 through 100.

prospectus 30 **Features information**

Use case goals	This use case is responsible for installing a GdM and possibly enrolling it in the operational network. It covers both the active network scenario and the deployment of the device when connection to the SAC is not possible, for example due to network coverage issues.
Level	Primary
Prerequisites	The field installer has a work order for installing the GdM. The installer has a portable device capable of managing the local connection to the GdM. The installer has the appropriate credentials to connect to the measurement group via the local port. The various security keys have been factory-installed in the measurement group. The SAC has received the GdM identification data and the keys from the manufacturer. Commands for the initial configuration parameters have been securely transferred to the portable device (handheld unit). The GdM has been set up (*for example*by inserting a SIM card) to register on the network.
Use case success condition	The instrument has been installed, and possibly enlisted.
Use Case Failure Condition	-
Primary roles	Field installer
Secondary roles	GdM SAC
Triggering cause	The installer installs the meter and optionally activates the network analysis function at the GdM level
Hypothesis	It is assumed that the field installer's requirements are limited to installing and configuring the GdM. The field installer is not responsible for troubleshooting network coverage issues for the device.
Security requirements	High
Category	Infrastructure management and maintenance
References	UNI/TS 11291-1:2013, point 4.3.8 UNI/TS 11291-6:2013, point 7.3 UNI/ TS 11291-10:2013, point 10.6.2

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The following scenarios are

considered: a) Main success scenario

10) This use case starts when the field installer mechanically assembles the GdM at the delivery point.

20) The installer connects the portable unit to the local port of the measuring group and establishes a secure connection. See point 10.6.2 of UNI/TS 11291-10:2013.

30) The installer programs the initial configuration parameters for the measurement unit. See UNI/TS 11291-6:2013, point 7.3.

40) The installer completes the initial programming and changes the GdM status to "normal". See UNI/TS 11291-6.2013, point 7.3.

Note 1 The initial schedule must include the date and time.

50) The installer activates the SAC connection test procedure on the GdM. The GdM must be placed in real operating conditions (e.g., the container door, if present, must be closed).

60) With the configurable delay, as provided by the manual communication parameters, the GdM activates the modem, registers itself to the network, and connects to the SAC.

Note 2 The delay is necessary to allow the GdM to restore its operating environmental conditions, e.g. closing the door of the niche in which it is installed.

70) The SAC sees a GdM, identifies it via the public client and sends an authenticated command to the GdM (at least containing a synchronization command).

80) The GdM receives the response from the SAC and provides information about the portable device (HHU). If necessary, the SAC forces the GdM out of the orphan state.

90) The field technician verifies that the GdM has adequate coverage and communicates with the SAC.

100) The SAC places the GdM on the list of enlisted counters. From this moment on, the GdM is enrolled in the SAC.

110) The device will start connecting to the SAC and send daily data and anything else requested by the SAC.

120) The use case ends.

Note 3 In case it is already known that there is no network coverage, the use case ends after step 40.

b) Alternative scenarios

- The installer is unable to establish a secure connection to the GdM.

  This alternative scenario is responsible for handling the situation where the field installer does not have the appropriate credentials to program the GdM via the local port

  - In step 20, the GdM determines that the connection establishment request fails the security check and an error is returned.

  - The use case ends.

- The connection is not possible due to a hardware problem.
  This alternative scenario is responsible for handling the situation where there is a problem that prevents the GdM from connecting to the network (e.g. missing or incorrectly mounted SIM).

  - In this case, the hardware problem must be resolved through maintenance or meter replacement. The procedure is similar to a new installation.

- The GdM is unable to establish a secure connection to the SAC.

  This alternative scenario is responsible for handling the situation where the SAC cannot be reached or, if reachable, the SAC is unable to produce an accepted command, for example because it does not have the relevant credentials.

  - The GdMs are still switched to normal mode and the registration process will evolve, as soon as a SAC has detected it and sent an accepted command.
- The GdM installation is programmed not to use the HHT.

  This use case assumes that steps 20 to 40 have been previously completed in the factory or warehouse, and consequently the GdM, upon arrival at the installation site, is already programmed and in normal state.

  In this case, steps 50 to 100 are only possible if the GdM provides a function to activate the connection to the SAC without the aid of HHT (e.g. via button).

  In any case, the initial or factory programming of the GdM must be such that step 110 is always possible, regardless of whether explicit enlistment is activated (steps 50 to 100) or not.

prospectus 31 **Additional Information**

Priority	See Appendix A
Expected performance	-
Frequency	Once during the first installation. NOTE: The requirements for the installation of the GdM are defined in UNI/TS 11291-6:2013.
Subordinating use case	-
Subordinate use cases	-

6.7.2 Enlistment

"Enlistment" means that the GdM is present in the database of a SAC, that all parameters necessary for normal operation have been configured in the GdM, and that the GdM has exchanged at least one valid message with the SAC.

Note 1 The GdM does not provide for any specific status that is changed by the enlistment process.

prospectus 32 **Features information**

Use case goals	This use case is responsible for the enrollment of GdMs within the SAC
Level	Primary
Prerequisites	The GdM has been installed. The PP4 communication network with the SAC is available. The SAC has received the GdM identification data and the keys from the manufacturer.
Use case success condition	The GdM is enrolled by a SAC, and all the parameters required for normal operation have been configured in the GdM. This means that the GdM is being properly managed by the SAC.
Use Case Failure Condition	The GdM fails to be recruited by a SAC
Primary roles	GdM SAC
Secondary roles	
Triggering cause	The GdM not yet enrolled makes a connection to the SAC.
Hypothesis	
Security requirements	High
Category	Infrastructure management and maintenance
References	UNI/TS 11291-1:2013, point 4.3.8 UNI/TS 11291-1:2013, point 7.3 (GdM states) UNI/ TS 11291-10:2013, point 10.9 (GdM states)

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The following scenarios are

considered: a) Main success scenario

10) This use case starts when the GdM activates the modem, registers to the network, and connects to the SAC via PP4.

Note 2 The operations in step 10 can be activated explicitly via HHU or button or performed autonomously by the GdM (programming for a periodic connection).

20) The SAC sees a GdM, and identifies it through information received from the GdM, e.g. via a push, or explicitly requested via the public client.

30) The SAC verifies that the new GdM is on its list of managed devices and, if necessary, sends at least one authenticated command to the GdM.

Note 3 The authenticated message may contain the dedicated command to exit the orphaned state.

40) The GdM accepts the command sent by the SAC.

50) The SAC places the GdM on the list of enlisted counters. From this moment, the GdM is enrolled in the SAC.

60) The SAC forms the configuration commands (safely) for all GdM parameters necessary for the operation of the GdM itself, if any.

70) The use case ends.

b) Alternative scenarios

- Enrollment takes place via local port connection.
- The authenticated command is transferred offline to the GdM via HHU using a pre-protected frame.
- In step 30 the SAC discovers that the GdM does not belong to its network (for example GdM from a different distributor)
  - The SAC does not send commands to the GdM
  - The GdM closes the communication due to timeout and, if configured, retries the connection
  - The use case ends in failure

prospectus 33 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements. See in particular point 4.3.8 of UNI/TS 11291-1:2013
Frequency	On an ad hoc basis (activated by an operator in the field or at a scheduled time for data sending).
Subordinating use case	-
Subordinate use cases	UC point 6.10.2 Writing

6.7.3 Orphan status management

prospectus 34 **Features information**

Use case goals	This use case is responsible for managing the orphaned state of the GdM. The orphan state corresponds to the condition that is reached after the GdM has not received a valid command1)for a programmable time.
Level	Primary
Prerequisites	The GdM has been installed.
Use case success condition	In the absence of communication with the SAC for a programmable time, the GdM goes into the orphan state, activates the S02 strategy and exits it only upon receiving the commanddedicated and valid by the SAC.
Use Case Failure Condition	
Primary roles	GdM SAC
Secondary roles	
Triggering cause	Missing for a programmable time / receipt of a valid command from the SAC.
Hypothesis	It is assumed that the factory-new meters are already in the orphan state.
Security requirements	High
Category	Infrastructure management and maintenance
References	UNI/TS 11291-12-2

1) According to UNITS 11291-10, a valid command is a correctly formatted command intended for the GdM that carries a non-empty application layer that is acceptable to the GdM and is correctly decrypted by the GdM and whose authenticity of the sender is verified.

The following scenarios are considered:

a) Main success scenario for entry into orphan status

10) This use case starts when the GdM has been successfully enrolled, for which is not in the orphan state.

20) The GdM does not receive any valid command for at least a programmable time (time-out).

30) The GdM becomes an orphan.

40) The GdM activates the S02 communication strategy, if appropriately configured. Otherwise, it remains in the current communication strategy.

50) The use case ends.

b) Main success scenario for exiting the orphan state

10) This use case starts when the GdM is in the orphan state.

20) The GdM receives the dedicated and valid command directed to him by the SAC.

30) The GdM leaves the orphan state.

40) The GdM activates the communication strategy S00 or S01 depending on the valve status.

50) The use case ends.

c) Alternative scenarios

- The GdM, which is not in the orphan state, receives at least one valid command directed to it by the SAC. The GdM remains in the non-orphan state and resets the programmable time (time-out).

prospectus 35 **Additional Information**

Priority	N/A
Expected performance	N/A
Frequency	Continues
Subordinating use case	-
Subordinate use cases	UC point 6.10.2 Writing

6.7.4 Radio Network Management

Radio network management requires a very complex set of use cases to implement all the processes and criteria dedicated to managing the communications infrastructure, in addition to those defined elsewhere in this specification (for example, installation and enrollment).

Such use cases are typically handled by the telephone operator and are therefore beyond the scope of this specification.

Note However, the use cases managed by the network operator are essential and contribute to achieving service level agreements (SLAs). For this reason, a certain level of communication service quality should be guaranteed based on the prescribed system SLAs.

Some network diagnostic activities are also performed using objects and functions available in the GdM. To enable interchangeability between network devices, these objects must be uniquely defined. This is achieved by defining specific Objects and Functions in the Application Object Database.

6.7.5 Battery Change

prospectus 36 **Features information**

Use case goals	This Use Case describes the GdM battery change operation. Because this activity can be potentially harmful, the system provides appropriate protection against tampering and/or fraud attempts. Therefore, any unauthorized access to the battery compartment will trigger an alarm and optionally (if configured) cause the valve to close.
Level	Primary
Prerequisites	The GdM has been correctly installed and is operational at the user's premises. Access to the installation site, at the end user's premises, has been managed and is secured. The battery is of the "replaceable" type.
Use case success condition	The battery has been replaced
Use Case Failure Condition	The battery was not replaced and optionally the valve closed (if the GdM was configured for this purpose).
Primary roles	SAC, GdM
Secondary roles	local operator, HHT
Triggering cause	Low battery information produced by the GdM and received by the SAC
Hypothesis	A secure connection between SAC and GdM has been established
Security requirements	High
Category	Infrastructure Management and Maintenance
References	UNI/TS 11291-10:2013, points 4.3 and 6.4.2

The following scenarios are

considered: a) Main success scenario

10) Following receipt of diagnostic information “battery life remaining <10%” from a GdM with unique identifier IDxx, the SAC issues an order: “battery replacement in GdM IDxx”

20) The SAC sends a message to the GdM IDxx to activate a maintenance session in a well-defined time interval, during which an operator will go to the installation site and carry out the battery replacement.

Note If required, a new maintenance key can also be programmed into the GdM.

30) The GdM acknowledges the received command and confirms it to the SAC.

40) During the above-mentioned time interval, the operator starts the maintenance session by explicitly associating with the I/M client (possibly with a pre-protected message that programs the maintenance key and the session duration) or by directly associating with the Management client with a pre-protected message.

50) The operator, with the aid of the HHT, enables the battery change (thus avoiding triggering the GdM tamper alarm).

60) The operator replaces the battery.

70) The operator, with the aid of the HHT, resets the battery life counter and the low battery event diagnostic and restores the normal operating mode.

80) The use case ends.

prospectus 37 **Related information**

Priority	See Appendix A
Performance Objectives	Refer to the relevant parts of UNI/TS 11291
Frequency	Once in the lifetime of the GdM.
Subordinating Use Case	Nobody
Subordinate Use Case	Nobody

6.7.6 GdM Replacement

prospectus 38 **Features information**

Use case goals	This Use Case is responsible for replacing a GdM
Level	Primary
Prerequisites	The installer has received the work order to replace the GdM. The installer is equipped with a portable terminal capable of connecting to the GdM's local optical port. The installer has the appropriate credentials and authorizations to connect to the GdM via the local optical port. The security keys have been factory programmed into the GdM. The SAC knows the GdM identifier and its keys (securely received from the manufacturer). The command set for the initial configuration of the parameters has been securely transferred to the handheld terminal. NOTE: The GdM to be replaced is itself a Smart Meter.
Use case success condition	The old GdM has been replaced by the new one and this one is operational, enrolled by the SAC.
Use Case Failure Condition	The old GdM has been replaced by the new one and this is operational, but the recruitment has failed.
Primary roles	Installer
Secondary roles	GdM SAC
Triggering cause	The installer begins the replacement at the user's premises.
Hypothesis	It is assumed that the necessary installed base management activities required for the replacement of a GdM have been carried out at the distributor's Management Center and that the SAC has been informed. This functional interaction between the Center and the SAC is outside the scope of Interchangeability. It is also assumed that the requirements to which the installer must comply are limited to the installation and configuration of the GdM. The installer is not required to perform analysis and troubleshooting in the event of a lack of network coverage.
Security requirements	High
Category	Infrastructure Management and Maintenance
References	UNI/TS 11291-1:2013, point 4.3.8 UNI/TS 11291-6:2013, point 7.3 UNI/ TS 11291-10:2013, point 10.6.2

The following scenarios are

considered: a) Main success scenario

10) This Use Case starts when the installer with his portable terminal connects to the optical port of the GdM to be removed and retrieves the following data, as far as possible:

- GdM identifier removed
- Data relating to the end of the billing period, for the last available periods
- Data relating to the current billing period
- All events present in the respective logs

20) Once this data has been recovered, the installer physically removes the old GdM and installs the new one at the return point.

30) The new GdM is installed and activated, following UC point 6.7.1: Installation and Activation of the GdM.

40) The SAC updates its list of enrolled GdMs to remove the ID of the removed GdM.

50) The use case ends.

b) Alternative scenario

- The GdM installation is programmed not to use the HHT.

  This use case assumes that steps 10 and 40 are performed offline at a later stage after the GdM has been removed.

- All possible alternative scenarios described in the Use Case at point 6.7.1 also apply.

prospectus 39 **Related information**

Priority	See Appendix A
Performance Objectives	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	Once in the useful life of the meter
Subordinating use case	Nobody
Subordinate Use Case	UC point 6.7.1 Installation and activation of the GdM

6.7.7 Change of Concession to a Third Party (another Distributor)

prospectus 40 **Features information**

Use case goals	This Use Case is responsible for managing the Change of Distributor (= Concession) of one or more GdMs enrolled by a SAC.
Level	Primary
Prerequisites	PP4 is available or the local port is (Local Concession Change). The concession change process is initiated by the old distributor (= Outgoing) who is responsible for the correct execution of the steps necessary to de-enroll each of the meters involved.
Use case success condition	The GdM involved is recruited by the SAC of the new Distributor (Substitute)
Use Case Failure Condition	The GdM involved is NOT enrolled by the SAC of the new Distributor (Substitute).
Primary roles	Outgoing SAC (remote UC) Incoming SAC (remote UC)
Secondary roles	HHT Operator Portable Terminal (UC from local) GdM
Triggering cause	Change of concession between one Distributor (outgoing) and another (incoming).
Hypothesis	It is assumed that a secure connection exists between SAC and the relevant GdMs. It is assumed that the Incoming Distributor knows the relevant service key for each GdM involved, i.e. the encryption key required for the secure execution of the concession change.
Security requirements	High. The critical aspect is the exchange of safety material.
Category	Infrastructure Management and Maintenance
References	UNI/TS 11291-10:2013

The following scenarios are considered: a) Main success scenario

10) This Use Case starts when the outgoing distributor starts the business transfer of meters from the outgoing SAC.

20) For each GdM to be transferred:

20A) The outgoing SAC sends a command to close the billing period to the GdM involved

20B) The GdM receives the command.

20C) The GdM processes the billing period close command which generates the related event which is stored in the event queue.

20D) The GdM transfers the end-of-period billing information to the SAC in accordance with the established communication strategy.

30) Once the outgoing SAC has received the end-of-billing-period reading messages from all the meters involved in the concession change, it can actually begin the transfer of those meters to the new network.

40) The outgoing Distributor, for each GdM involved, provides the incoming distributor, in a safe manner as foreseen in UNI/TS 11291-10:2013, and with the methods set out in UC point 6.8.1 Supply of Keys, at least the identifier and the service key.

50) The above data is stored in the incoming SAC.

60) The outgoing SAC reprograms the GdMs so that they point to the incoming SAC starting from the subsequent communications.

70) The SAC of the incoming Distributor begins to receive connection requests from the GdMs subject to the transfer.

80) The SAC identifies each new GdM that appears on the PP4 network and, if recognized based on the list received from the outgoing Distributor, inserts it into the list of GdMs supported by the system.

90) The SAC prepares secure commands in order to configure the operating key of each affected GdM with a new value (see UC point 6.9.2 Updating keys) using the current security material as received from the outgoing Distributor.

100) For each GdM involved and recognized by the incoming SAC, that SAC begins the enrollment phase.

110) The SAC updates the list of enrolled GdMs to indicate that all new GdMs are enrolled in the system.

120) The use case ends.

b) Alternative scenarios

- One or more of the Counters involved fail to complete:

a) key update;

b) the enrollment process within a predefined time frame (e.g. 2 weeks).

Previous failure conditions could be resolved by local intervention. **Related information**

prospectus 41

Priority	See Appendix A
Performance goals	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	No more than once in a year.
Subordinate Use Case	-
Subordinate Use Case	Point 6.7.2 Enrollment Point 6.8.2 Key Update Point 6.8.1 Key Supply

6.8 Security Management

6.8.1 Provision of keys

Security keys must be transferred securely to the various system actors.

Most hypothesized keys are generated at the factory by the manufacturer and therefore must be sent to the relevant users in a secure manner. UNI/TS 11291-10:2013 defines different scenarios that may or may not include an independent third-party guarantor.

Note 1 The role of guarantor authority could also be covered by a Distributor or a Manufacturer, in which case the limitations on the security architecture described in UNI/TS 11291-10:2013 apply.

This Use Case is not intended to specify the internal processes that manufacturers or distributors adopt to ensure the security of keys before transmitting them or during and after the key reception phase.

As an integral part of this Use Case, the key exchange format and the related key exchange schematics, as well as the file formats to be used, are described in the appendices to this document.

Three types and related key exchange formats are defined:

- Transfer of the set of keys from the manufacturer to the distributor.

- Transfer of the service key to the third-party guarantor.

- Transfer of maintenance key to installers/maintenance technicians (exported from SAC).

Note 2 This context also includes the set of pre-protected messages that the SAC generates in order not to expose the meter keys on the maintenance workers' portable terminals (HHT).

The same concepts apply in all other cases of key exchange permitted in the system, between primary and secondary actors, as defined in Table 44. To facilitate understanding, typical cases of key transfer between Manufacturers and Distributors are identified below.

prospectus 42 **Key exchange**

Destination	**M**	**THE**	**U**	**S**	**G**
Source	Manufacturer	Installer	Distributor	SAC	Guarantor Authority
M Manufacturer			X		X
THE Installer					
U Distributor		X	X	X	
S SAC					
G Guarantor Authority					

NOTE
M sends the following keys to U: KEYT / KEYC
M sends the following keys to G: KEYS
U sends the following keys to I: KEYT U
sends the following keys to S: KEYC
If the guarantor authority is not available, then one of M and U will act as G.
The U--U transfer applies only to the UC use case point 6.7.7 “Change of Concession” in the case where the role of the Guarantor Authority is carried out by the Distributor and consists of the transfer of at least the KEYS.

prospectus 43 **Features information**

Use case goals	This use case is responsible for distributing symmetric keys between the GdM Manufacturer and the Distributor and/or the Guarantee Authority in the form of a key transport file (EKSF). These keys are used to secure communication between the GdM and the SAC, including installation and maintenance of the GdM once it has been deployed in the field.
Level	Primary
Prerequisites	The GdM was equipped with a set of symmetrical keys during the manufacturing process. The following general requirements for both the Manufacturer and the Gas Distributor's sites are met: - A clear separation in key usage must be maintained. Each key (whether symmetric or asymmetric) has a single, dedicated function. - A cryptographically secure pseudorandom number generator (RNG) and a random number as a seed must be used to generate asymmetric keys and initialization vectors (IVs). The RNG must comply with either NIST SP 800-90 1 (preferred) or FIPS PUB 186-2. - Certificates cannot be self-signed. Certificates must be signed by a registered certificate authority (CA). The GdM Manufacturer has generated an asymmetric key pair (PA_EKSF_SIGN_PRIV and PA_EKSF_SIGN_PUB) and a corresponding certificate (PA_EKSF_SIGN_CERT). EKSF is signed by PA_EKSF_SIGN_PRIV: - asymmetric key pair: - RSA with a key length of at least 2,048 bits - or ECC-256 with NIST secp256r1 curve - certificate format: X509.v3; certificate file extension must be der, pem, or pkcs12 - hash function: SHA-256 - Digital signature algorithm: - RSA - or EC-DSA The Distributor has generated an asymmetric key pair (SA_EKSF_CRYPT_PRIV and SA_EKSF_CRYPT_PUB) and a corresponding certificate (SA_EKSF_CRYPT_CERT): - asymmetric key pair: - RSA with a key length of at least 2,048 bits - or ECC-256 with NIST secp256r1 curve - certificate format: X509.v3; certificate file extension must be der, pem, or pkcs12 - hash function: SHA-256 - Digital signature algorithm: - RSA - or EC-DSA The Manufacturer and the Distributor have exchanged their certificates (PA_EKSF_SIGN_CERT and SA_EKSF_CRYPT_CERT), for example, in person or via certified email (PEC). The GdM Manufacturer and the Distributor have checked the certificate revocation list. The GdM Manufacturer and the Distributor have verified the authenticity of the exchanged certificates via a trusted external channel (for example, by a phone call between known individuals to verify the key fingerprint or by sending a signed email). The GdM Manufacturer and the Distributor have extracted the public key from the received certificate. NOTE: The same concepts and rules apply to the transfer from the manufacturer to the Guarantee Authority, the only difference being the file format.

prospectus 43 **Features information**(Continues)

Use case success condition	The EKSF with the keys is received by the Distributor and the authenticity verification and decryption of the EKSF are successfully performed by the Distributor.
Use Case Failure Condition	The EKSF with the keys is not received by the Distributor or the authenticity verification and decryption of the EKSF fail.
Primary roles	According to this prospectus
Secondary roles	According to this prospectus
Triggering cause	The Manufacturer has produced the GdM and must send it to the distributor
Hypothesis	
Security requirements	
Category	Security Management
References	UNI/TS 11291-10:2013, point 10.6: Key distribution must be in accordance with NIST SP800-57[1]
1) NIST Special Publication 800-57: Recommendation for Key Management.	

The following scenarios are

considered: a) Main success scenario

10) This use case starts when the Manufacturer has produced the GdMs and has to send them to the distributor.

20) The GdM Manufacturer encrypts all symmetric keys (each key separately) of all GdMs produced using SA_EKSF_CRYPT_PUB and with one of the following mechanisms:

- key encapsulation (KEM) according to RFC 5990[1]and the PKCS#1 padding scheme

- or ECIES-KEM[2](Elliptic Curve Integrated Encryption Scheme KEM).

30) The GdM Manufacturer writes the encrypted symmetric keys into the EKSF.

The format of the EKSF must be as specified in Appendix D.

40) The GdM Manufacturer signs the EKSF with PA_EKSF_SIGN_PRIV using the hash function and digital signature algorithm as specified in the previous preconditions.

50) The GdM Manufacturer transmits the EKSF to the Distributor and/or the Guarantor Authority on a personal basis or by email.

60) The Distributor verifies the EKSF signature using PA_EKSF_SIGN_PUB.

70) If the signature verification is successful, the Distributor decrypts all encrypted keys of the equipment using SA_EKSF_CRYPT_PRIV.

80) The use case ends.

b) Alternative scenarios:

- Failure to create EKSF (step 20 and/or 30).
- EKSF signing failure (step 40).
- Transmission failure from the equipment manufacturer to the EKSF Distributor (step 50).
- EKSF signature verification failure (step 60).
- Failure to decrypt keys (step 70).

In all these cases, the entire process must be re-initialized.

1) RFC 5990: "Use of the RSA-KEM Key Transport Algorithm in the Cryptographic Message Syntax (CMS)", September 2010.

2) http://www.w3.org/TR/xmlsec-generic-hybrid/#sec-ecies-kem.

prospectus 44 **Related information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	This use case occurs for each delivery batch
Subordinating use case	Nobody
Subordinate use case	Nobody

6.8.2 Key Update

This use case is divided into different sub-cases depending on the type of key to be updated.

6.8.2.1 Updating the GdM operating key

prospectus 45 **Features information**

Use case goals	This use case deals with the remote update of the operating key (KEYC) used for encryption and authentication of messages exchanged between the SAC and the GdM.
Level	Primary
Prerequisites	The GdM is equipped with an operating key (KEYC). The GdM has been installed and is functional. Access via PP4 is available.
Use case success condition	The operating key is activated
Use Case Failure Condition	The operating key is not activated
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC sends the request to change the operating key of a GdM
Hypothesis	The SAC is connected and authenticated KEYC is associated with an FC_c (Frame Counter) which is needed to uniquely identify each message authenticated/encrypted with KEYC. KEYC_old is the GdM's operating key; KEYC_new is the updated value of the operating key;
Security requirements	High: The message to activate the key must be encrypted and authenticated
Category	Key Update
References	UNI/TS 11291-10:2013

The following scenarios are

considered: a) Main success scenario

10) This use case starts when the SAC sends a request to the GdM to update the operating key (KEYC); the request containing the value of the new KEYC must be encrypted and authenticated with KEYC_old;

20) Until it receives the response from the GdM, the SAC maintains the values of both keys (KEYC_new and KEYC_old).

30) The GdM sends confirmation to the SAC, activates the new KEYC value and sets the FC associated with the KEYC to zero.

40) The SAC permanently deletes the KEYC_old and sets the associated FC to zero.

60) The use case ends.

b) Alternative scenarios

30) The confirmation sent by the GdM does not reach the SAC

- the SAC activates the operations necessary to understand whether the key change has been carried out or not

30) The GdM responds with a negative confirmation reporting the error conditions

- the SAC repeats the request

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prospectus 46 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	If necessary
Subordinating use case	Nobody
Subordinate use case	UC point 6.11.2: Basic writing

6.8.2.2 Remote activation of the GdM maintenance key

prospectus 47 **Features information**

Use case goals	This use case deals with the remote activation of the maintenance key (KEYT) used for encryption and authentication of messages exchanged between the HHT and the GdM.
Level	Primary
Prerequisites	The GdM is equipped with an operating key (KEYC). The GdM has been installed and is functional. Access via PP4 is available.
Use case success condition	The maintenance key is activated
Use Case Failure Condition	The maintenance key is not activated
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC sends a request to activate a GdM's maintenance key; the request also includes the types of activities permitted to the maintainer.
Hypothesis	The SAC is connected and authenticated KEYT is associated with an FC_t (Frame Counter) which is necessary to uniquely identify each message. KEYT is the maintenance key that must be activated; KEYC is the GdM operating key.
Security requirements	High: The key activation message must be encrypted and authenticated
Category	Key Update
References	UNI/TS 11291-10:2013

The following scenarios are considered: a) Main success scenario

10) This use case starts when the SAC sends a request to the GdM to activate the maintenance key (KEYT); the request must be encrypted and authenticated with the KEYC; the request must indicate:

- The date (date_start expressed as year, month, day, hour) from when it is possible to establish the installer/maintenance technician association to which the key refers
- The duration (starting from date_start) after which it will no longer be possible to establish an installer/maintenance association
- The activities permitted to the maintenance technician, namely:
    - Parameter changes, i.e. any SET or ACTION on any object
    - Resetting the metrological event log
    - Change to metrologically relevant parameters
    - Valve Management
    - Changing the Battery
    - Clock synchronization
    - Database reset
    - Parameter changes, i.e. any SET or ACTION on any remote communication management object

The detailed list of objects linked to each of the above permissions can be found in UNI/TS 11291-12-2.

20) The GdM receives and sends the confirmation to the SAC

30) At date_start, the GdM allows the activation of an association with the maintenance client using the KEYT

40) Upon expiration of the validity period, the GdM deactivates any association in progress with the maintenance client (subsequent messages encrypted and/or authenticated with the KEYT are rejected) and prevents the activation of further associations.

50) The use case ends.

b) Alternative scenarios

20) The confirmation sent by the GdM does not reach the SAC

- the SAC activates the operations necessary to understand whether the key change has been carried out or not

20) The GdM responds with a negative confirmation reporting the error conditions

- the SAC repeats the request

prospectus 48 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	If necessary
Subordinating use case	Nobody
Subordinate use case	UC point 6.11.2: Basic writing

6.8.2.3 Local activation of the GdM maintenance key

prospectus 49 **Features information**

Use case goals	This use case deals with the activation of the maintenance key (KEYT) used for encryption and authentication of messages exchanged between the HHT and the GdM when the key cannot be activated directly from the SAC remotely.
Level	Primary
Prerequisites	The GdM is equipped with an operating key. The GdM has been installed and is operational. Access via PP4 is not available.
Use case success condition	Maintenance key is activated
Use Case Failure Condition	Maintenance key is not activated
Primary roles	SAC HHT
Secondary roles	GdM
Triggering cause	The SAC sends a request to activate a GdM's maintenance key; the request also specifies the types of activities permitted to the maintainer.
Hypothesis	The SAC is not connected The maintenance key is associated with an FC needed to uniquely identify each message. KEYT is the maintenance key that must be activated. KEYC is the operating key of the GdM.
Security requirements	High: The message to activate the key must be encrypted and authenticated
Category	Key Update
References	UNI/TS 11291-10:2013

The following scenarios are

considered: a) Main success scenario

10) This use case begins when the SAC prepares a request to activate the maintenance key (KEYT) of a GdM; the request must be encrypted and authenticated with the KEYC; the request must specify:

- The date (date_start expressed as year, month, day, hour) from when it is possible to establish the installer/maintenance technician association to which the key refers
- The duration (starting from date_start) after which it will no longer be possible to establish an installer/maintenance association
- Activities permitted to the maintenance technician such as:
  - Parameter changes, i.e. any SET or ACTION on any object
  - Resetting the metrological event log
  - Change to metrologically relevant parameters
  - Valve Management
  - Changing the Battery
  - Clock synchronization
  - Database reset
  - Parameter changes, i.e. any SET or ACTION on any remote communication management object

The detailed list of permits can be found in UNI/TS 11291-11-2.

20) The request is sent to the HHT as a pre-encrypted message

30) The HHT is connected and authenticated by the GdM

40) The HHT transmits the maintenance key activation request to the GdM

50) The GdM receives, confirms to the HHT and activates the possibility to establish an association with the KEYT maintenance client

60) Upon expiration of the validity period, the GdM deactivates any association in progress with the maintenance client (subsequent messages encrypted and/or authenticated with the KEYT are rejected) and prevents the activation of further associations.

70) The use case ends.

b) Alternative scenarios

- 40) The GdM does not confirm
  - The HHT retries the command for a limited time within the limits set by the maximum number of “replays” of a command possible
- 40) The GdM or Repeater responds with a negative confirmation reporting the error conditions
  - The HHT stores the response and stops all further communication

prospectus 50 **Related information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	If necessary
Subordinate use cases	Nobody
Subordinate use cases	UC point 6.11.2: Basic writing

6.9 Firmware Update

6.9.1 Updating the Firmware of a GdM

The GdM firmware consists of one or more components.

prospectus 51 **Features information**

Use case goals	This use case describes the remote and local deployment of the new firmware version image for the GdM.
Level	Primary
Prerequisites	PP4 connection is active. A new image of a GdM firmware component has been released by the manufacturer. The SAC recognizes that a GdM firmware component needs to be updated. The SAC determines which GdM should receive the image. The authentication key (KDL) has been generated. The SAC has securely received the signature (hash tag) relating to the image of the component to be transferred.
Use case success condition	The image of the new firmware component has been downloaded and subsequently activated successfully on the subject GdM.
Use Case Failure Condition	The image of the new firmware component is not activated on the GdM in question.
Primary roles	SAC
Secondary roles	Local Device HHT (Local Firmware Component Update) GdM
Triggering cause	The SAC issues the request to update the GdM firmware component.
Hypothesis	Scheduling the software transfer is the responsibility of the SAC.
Security requirements	High (critical command); firmware transfer must be initiated from the administrator role.
Category	Software update
References	UNI/TS 11291-1:2013, point 4.3.6

The following scenarios are

considered: a) Main success scenario

10) This use case starts when the SAC sends a request to update a firmware component to the reference GdM. This request contains:

- enabling image transfer
- initialization of the image transfer which must contain at least the authenticity key, the signature and the date of activation of the image

this request can arrive via PP4 directly from the SAC or via the HHT as a pre-protected message, and must always be encrypted with the operating key;

20) The GdM receives the command to transfer an image block from the SAC or HHT and confirms receipt;

30) The SAC or HHT sends the software update to the GdM in question; the SAC or HHT can request information on which block was last successfully transferred and resume the update from that;

40) The GdM receives the firmware component update; the GdM successfully accepts the new block only in the image transfer started state;

41) The GdM carries out the verification of integrity and authenticity of the image only when it has been received in its entirety;

50) The GdM schedules the activation of the firmware component on the activation date, initialized and configured via the initialization command, only when its integrity and authenticity have been successfully verified;

60) The GdM makes available to the SAC or HHT the information that the provided image has been successfully verified

70) On the scheduled date, the GdM verifies the integrity of the firmware component and activates the update; upon completion of the update, the GdM makes the information regarding the correct activation available to the SAC or HHT;

80) The use case ends.

b) Alternative scenarios

- Image transfer completion exceeds activation date
  - Verification will be performed upon completion of the transfer as per step 40;
  - Upon successful verification, the image is activated immediately;
- Firmware component image fails authenticity verification
  - At step 40, the GdM fails to verify the integrity and authenticity of the firmware component image;
  - The GdM generates an event in the metrology log and makes the information available to the SAC or HHT that the firmware component image was “corrupted”;
  - Upon receiving further firmware blocks, the GdM responds with an error; this condition is resolved by explicitly reinitializing the process by the SAC;
  - The use case ends.
- Firmware component image is not activated
  - At step 70, the GdM fails to activate the image;
  - The GdM generates an event in the metrological register and makes the information available to the SAC or HHT that the image activation was not successful;
  - The use case ends.

prospectus 52 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	Firmware updates occur as needed.
Subordinating use case	Nobody
Subordinate use cases	Nobody

6.10 Basic use cases

The following read and write use case scenarios are described assuming that only one instance of the requested service is managed for each communication session. Multiple instances can be present in the same session (e.g., a read followed by a write), in which case all instances are considered to be within the same communication session and mutually authenticated.

The same use cases are described for the scenario involving a mutual authentication step between the communication actors. The same scenarios can also be used without this step (for example, because the protocol's pre-established association mechanism is used). In this case, step 30 must be omitted from both use cases. Furthermore, in this case, the session (step 70) cannot be terminated by the SAC using the association release mechanisms.

In the following scenarios the concept of*application timeout*All application timeouts must be configured appropriately for the communication system used, taking into account not only the latencies of the communication channel, but also the delays introduced by all communication actors, including the SAC.

6.10.1 Reading

prospectus 53 **Features information**

Use case goals	This use case is responsible for remote reading of data from the GdM (via PP4)
Level	Primary
Prerequisites	The GdM has been installed and is operational at the customer's premises. Access via PP4 is available.
Use case success condition	The data is read by the GdM
Use Case Failure Condition	The data is not read by the GdM
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC issues the request to read data from a GdM
Hypothesis	It is assumed that a secure connection is established between the SAC and the GdM as part of this use case.
Security requirements	Variable (depending on the content to be read by the GdM); data must be protected by default.
Category	Basic services
References	-

The following scenarios are considered: a) Main success scenario

10) This use case starts when the GdM establishes a communication session with the SAC at defined and programmable times.

20) The SAC identifies the caller through information, such as the unique identifier, which the GdM makes accessible without any protection.

Note 1 If the GdM sends a push message upon successful connection, it must contain the information required for this identification.

30) The SAC and the GdM proceed, if required by the client used, to mutual authentication using the mechanisms made available by the communication protocol (for example HLS in the case of DLMS).

Note 2 If you are using a predefined association, this step is not necessary.

40) The SAC forwards the data reading request to the GdM.

50) The GdM receives, validates and processes the request for *Reading.*

60) The GdM sends the requested data to the SAC.

70) The SAC releases the connection with the GdM.

80) The use case ends.

b) Alternative scenarios

- The PP4 communication channel is not available (step 10 fails and the use case ends)
- The SAC does not identify the caller because, for example, it does not belong to the list of managed GdMs (step 20 fails and the use case ends).
- Authentication fails (step 30: use case ends)
- Validation of the read request fails or the tool is unable to process the read request. *Reading* (step 40 or step 50 fails)
- The requested data are not available for reading depending on the software version, the type of GdM, the access rights
- The data read request does not pass integrity and security checks
- The SAC does not receive the response containing the requested data (step 60 fails: the SAC can retry the read operation a preset number of times).

- The SAC does not release the connection to the GdM. This does not represent a failure condition, as the GdM closes the connection due to an (application) timeout, and the use case therefore ends successfully (step 70).

prospectus 54 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	Generic use case.
Subordinating use case	Read basic profile, read clock, read event log, read billing data, read tariff data, read tariff program, read consumption profiles, read data, read status
Subordinate use cases	Nobody

6.10.2 Writing

prospectus 55 **Features information**

Use case goals	This use case is responsible for remotely writing data to the GdM (via PP4)
Level	Primary
Prerequisites	The GdM has been installed and is operational at the customer's premises. Access via PP4 is available.
Use case success condition	The data is written in the GdM
Use Case Failure Condition	The data is not written in the GdM
Primary roles	SAC
Secondary roles	GdM
Triggering cause	The SAC issues the request to write data to a GdM
Hypothesis	It is assumed that a secure connection is established between the SAC and the GdM as part of this use case.
Security requirements	Variable (depending on the content to be written by the GdM); write commands must always be at least authenticated by default.
Category	Basic services
References	-

The following scenarios are considered: a) Main success scenario

10) This use case starts when the GdM establishes a communication session with the SAC at defined and programmable times.

20) The SAC identifies the caller through information, such as the unique identifier, which the GdM makes accessible without any protection.

Note 1 If the GdM sends a push message upon successful connection, it must contain the information required for this identification.

30) The SAC and the GdM proceed, if required by the client used, to mutual authentication using the mechanisms made available by the communication protocol (for example HLS in the case of DLMS).

Note 2 If you are using a predefined association, this step is not necessary.

40) The SAC forwards the data writing request to the GdM.

50) The GdM receives, validates and processes the request for *writing*.

60) The GdM sends a confirmation to the SAC.

70) The SAC releases the connection with the GdM.

80) The use case ends.

b) Alternative scenarios

- The PP4 communication channel is not available (step 10 fails and the use case ends)
- The SAC does not identify the caller because, for example, it does not belong to the list of managed GdMs (step 20 fails and the use case ends).
- Authentication fails (step 30: use case ends)
- Validation of the write request fails or the tool is unable to process the write request.*data writing*(step 40 or step 50 fails)
  - The data and/or request are not available for writing depending on the firmware version, GdM type, access rights
  - the data write request fails integrity and security checks
- The SAC does not receive write confirmation (step 60 fails: the SAC can retry the write operation a preset number of times).
- The SAC does not release the connection to the GdM. This does not represent a failure condition, as the GdM closes the connection due to an (application) timeout, and the use case therefore ends successfully (step 70).

prospectus 56 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	Generic use case.
Subordinating use case	Setting clock parameters, clock and synchronization settings, setting end-of-billing parameters, setting the tariff plan, interrupting supply, reactivating supply, enrollment, updating keys, updating FW
Subordinate use cases	Nobody

6.10.3 Push

prospectus 57 **Features information**

Use case goals	This use case is responsible for sending a predefined sequence of data on a regular basis (via PP4). This data should be that which is regularly needed by the SAC (e.g., daily measurements, GdM status).
Level	Primary
Prerequisites	The GdM has been installed and is operational at the customer's premises. Access via PP4 is available.
Use case success condition	The data is received from the SAC.
Use Case Failure Condition	Data is not received from the SAC.
Primary roles	GdM
Secondary roles	SAC
Triggering cause	Timed activation
Hypothesis	It is assumed that a push strategy (timing and data sequence) has been programmed into the GdM.
Security requirements	The data must not be decryptable or alterable by an external agent: the data is at least encrypted by default.
Category	Basic services
References	-

The following scenarios are

considered: a) Main success scenario

10) The GdM timer clock activates the connection sequence.

20) the GdM establishes a communication session with the SAC.

30) The GdM sends the pre-established data sequence via PP4, or waits for a request from the SAC.

40) The SAC receives the data sent by the GdM or requests the necessary data from the GDM.

50) The GdM remains listening for a predefined period of time for a possible request from the SAC (therefore UC point 6.11.1 Reading or UC point 6.11.2 Writing will follow).

60) The SAC releases the connection with the GdM using a dedicated command or the session ends due to a timeout (application).

70) The use case ends.

b) Alternative scenarios (except those already indicated in points 6.11.1 and 6.11.2)

- The PP4 communication channel is unavailable (step 20 fails): the GdM triggers new deferred retries as scheduled. When the scheduled number of retries expires, the use case ends.
- The GdM does not receive messages from the SAC (step 50 fails): the GdM possibly triggers new deferred retries as scheduled. When the scheduled number of retries expires, the use case ends.

For construction details, please refer to the objects defined by UNI/TS 11291-12-2 and the timing diagrams reported in UNI/TS 11291-12-4.

prospectus 58 **Additional Information**

Priority	See Appendix A
Expected performance	Refer to the relevant parts of UNI/TS 11291 for performance requirements.
Frequency	as scheduled depending on the defined SLAs
Subordinating use case	Setting and synchronizing the clock, setting the clock parameters, reading the end-of-billing data.
Subordinate use cases	Writing, reading

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APPENDIX TO DEFINING SERVICE LEVELS FOR USE CASES AND OTHER REFERENCE PARAMETERS

(regulations)

A.1

General information

The following table A.1 defines the service level agreement for different use cases.

To facilitate interpretation, SLAs have been grouped into five levels of importance, which require different percentage limits. They represent the fraction of the population that meets the service requirement within the defined time.

prospectus A.1 **ALS Classification**

Grade	IMPORTANCE	Percentage over time prescribed	Notes / Examples
1	HIGH	**90% - 94%**	Valve Management
2	MEDIUM-HIGH	**85% - 90%**	Reading billing data
3	MEDIUM	**78% - 85%**	Diagnostics/Alarms
4	MEDIUM_LOW	**68% - 78%**	Reading event log
5	BASS	**58% - 68%**	Reading the clock
NOTE "SLA" indicates the maximum prescribed time that the system must satisfy in the realization of the service in question.			

These values are referred to in UNI/TS 11291-1:2013 and are considered normative. The following table A.2 defines the different use cases.

prospectus A.2 **Use cases**

	Type	ALS	Percentage in definite time	Frequency in duration operational	Notes / References
6.2	Synchronization				
6.2.2	Setting and synchronizing the clock	3 d	MEDIUM_LOW	Whenever possible	Residual error and synchronization limits from UNI/TS 11291-1:2013, point 4.3.1
6.2.3	Setting the clock parameters	7 d	BASS	Once a year	
6.2.4	Reading the clock	N/A	BASS	Included in other information exchanges	
6.3	Detection and reporting of anomalies				
6.3.1	Fraud detection and reporting				
6.3.2	Alarm management from GdMs		MEDIUM-HIGH		
6.3.3	Reading diagnostic data and alarms	3 d	MEDIUM-HIGH1)/ MEDIUM	Referred to SLA (3 d)	UNI/TS 11291-1:2013, point 4.3.9.
6.3.4	Reading the event log	7 d	MEDIUM_LOW / BASS	If alarm present	
6.3.5	Local event log deletion	7 d	MEDIUM_LOW / BASS	Rarely/only for counters with register full events	
6.4	Dynamic multi-tariff capacity				
6.4.1	Configuring End of Billing (EOB) parameters	7 d	BASS	Once a year	
6.4.2	Reading the end-of-billing data	5 d	MEDIUM-HIGH	Monthly + in case of commercial event	Interval period
6.4.3	Reading the current totalizers	7 d	MEDIUM_LOW	If necessary	UNI/TS 11291-1:2013, point 4.3.7.

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table A.2 **Use cases**(Continues)

	Type	ALS	Percentage in definite time	Frequency in duration operational	Notes / References
6.4.4	Reading End of Billing (EOB) parameters	7 d	MEDIUM_LOW	Rarely	
6.4.5	Reading the tariff plan	7 d	BASS	Rarely	
6.4.6	Tariff plan programming	7 d	MEDIUM_LOW	Twice/year	UNI/TS 11291-1:2013, point 4.3.3.
6.5	Measurement and load profiles				
6.5.1	Reading load profile data	7 d	MEDIUM-HIGH	Every three days2)	UNI/TS 11291-1:2013, point 4.3.7.
6.6	Supply Management				
6.6.1	Supply interruption	7 d	HIGH	Twice a year	UNI/TS 11291-1:2013, point 4.3.4.
6.6.2	Reactivation of supply	24 hours	HIGH	Twice a year	
6.6.3	Reading the status	3 d	HIGH	Referred to SLA (3 d)	
6.7	Infrastructure Management and Maintenance				
6.7.1	Installation and activation of the GdM	3 d	HIGH	Once	UNI/TS 11291-1:2013, point 4.3.8.
6.7.2	Enlistment	3 d	MEDIUM-HIGH	Once	See above
6.7.3	Orphan status management	See description in the text of this document			
6.7.4	Radio Network Management	7 d	HIGH	Always operational	Any changes in the network must be made within 7 days. UNI/TS 11291-1:2013, point 4.3.8.
6.7.5	Battery change	7 d	HIGH	Once a year	UNI/TS 11291-10:2013, point 6.4
		NOTE SLA refers to the GdM enabling the battery change and not to the entire battery replacement process			
6.7.6	GdM Replacement	7 d	MEDIUM	Once	See above
6.7.7	Change of concession to a third party	15 d	MEDIUM_LOW	Once	
6.8	Security Management				
6.8.1	Provision of keys		MEDIUM	Out of scope SLA	
6.8.2	Key Update	7 d	MEDIUM	Twice/year	
6.9	Firmware update				
6.9.1	Updating the firmware of a GdM	45 d	HIGH	Twice/useful life	UNI/TS 11291-1:2013, point 4.3.4.
6.10	Basic use cases				
6.10.1	Reading		N/A	N/A	The classification of basic use cases is highly dependent on the subordinate use case.
6.10.2	Writing		N/A	N/A	
6.10.3	Push		N/A	N/A	

NOTE
The "SLA" column defines a time interval. This interval is delimited by a start and end time.
"Frequency" indicates how frequently, or how many times, the services in question are performed during the useful life. For further details, refer to the specifications in the "Notes / References" column.
1) For these use cases: 6.3.2, 6.3.3, 6.3.4, a higher level of importance applies in the case of anti-fraud features.
2) Refers to daily totals.

As defined in UNI/TS 11291-1:2013, the initial time is when a command is set by an operator in the SAC (e.g., interruption of supply, configuration of a new tariff plan, etc.) or when a triggering event occurs (e.g., matching the EOB date and time, the occurrence of an alarm). The final time of the SLA interval is determined by the moment the information (success/failure) becomes available in the SAC.

A.2 Communication strategies

Communication strategies must be appropriate for achieving SLAs. It is expressly noted that the times indicated in Table A.2 are not necessarily equal to the maximum period that must elapse between two successive connections.

Furthermore, for greater service availability, it is recommended to use dynamic connection strategies based on system status, with the aim of reducing communication frequency when a less stringent SLA applies. An example of this strategy can be found in valve management, where the time limits for closing and reopening operations differ significantly. This allows the connection frequency to be modulated based on the valve's status.

prospectus A.3 **Reference communication strategies (default)**

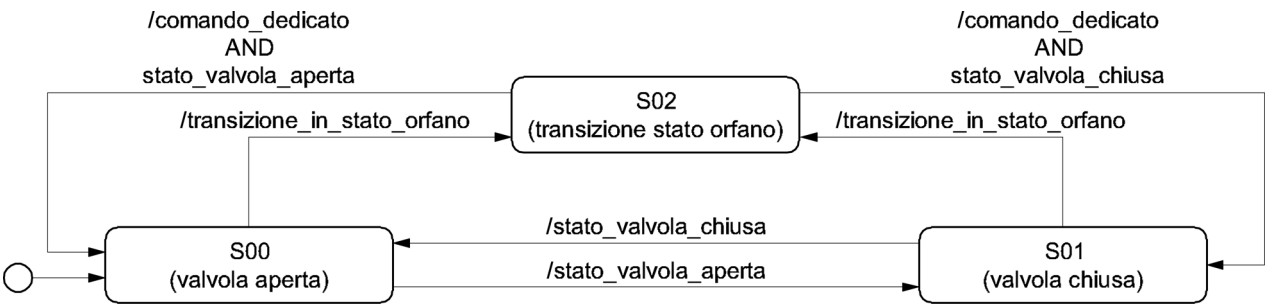
Strategy	Description	Frequency	Notes
S00	Normal Operation	1 communication every 3 days	
S01	User Disconnected	1 communication per day	To allow for rapid reactivation of the supply and to highlight any valve leaks
S02	Orphan GdM	Different communication frequency (e.g. once every 7 days)	

The transition from strategy S00 to S01 and vice versa occurs automatically, based on the valve status.

The S02 communication strategy is activated when the GdM is in the "orphan" state, i.e. not communicating with the SAC for more than a set time; the S02 strategy has priority over the S00 and S01 strategies.

Figure A.1 highlights the transitions between the different communication strategies.

figure A.1 **State diagram of communication strategies**



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A.3 Other parameters

The reference parameters in Tables A.4 and A.5, together with the reference service levels defined in this Appendix, must be used by manufacturers in defining the battery life of the GdM. The tables provide various application scenarios, and in particular a basic scenario, which represents the default communication strategies, a limit scenario that represents the case in which daily communication is required for prolonged periods (for example, in the case of repeated defaults), and an intermediate scenario that considers the case of communication every two days in the event of an open valve. The communication times are defined both with reference to the most efficient spontaneous communication strategy (Figure 2 of UNI/TS 11291-12-4), which constitutes the normal case, and in the case in which there is no spontaneous communication (Figure 5 of UNI/TS 11291-12-4).

prospectus A.4 **Reference parameters for GdM with GPRS communication**

GdM	Benchmarks			Notes
Number of communications with valve open	**Once every three days**	Once every two days	Once a day	Two end-of-billing communications must also be added, once a month.
Number of communications with valve closed	Once a day			
Duration of registration on the telecommunications network	30 s			Includes where applicable the time required to obtain the IP channel. NOTE This excludes the very first connection of a SIM in a new installation.
Duration of remote data communication (with spontaneous)	30 s	25 s	20 s	The time is averaged considering an additional number of communications related to data recovery in case of intermittent communication situations
Duration of remote data communication (without spontaneous)	40 s	35 s	25 s	
Transmission power	class 8 P_{tx}=30 dBm			GPRS as a reference
Local display operation	10 min / month			
Local interface operation	20 min / year			
Firmware update	3 in the useful life of the GdM			Communication times for FW update are not included in the above times.
Valve activation	30 cycles in the useful life			A cycle means an opening and a closing
Minimum battery life (with spontaneous)	**TO**	**B**	**C**	
Minimum battery life (without spontaneous)	**D**	**AND**	**$Tstd_{batGPRS}$**	

Tstd battery life values$_{batGPRS}$, A, B, C, D and E must be declared by the manufacturer with reference to the parameters reported in table A.4.

Values A, B, C, D and E must be greater than $Tstd_{batGPRS}$.

table A.5 **Benchmarks for GdM with NB-IoT communication**

GdM	Benchmarks			Notes
Number of communications with valve open	**Once every three days**	Once every two days	Once a day	Two end-of-billing communications must also be added, once a month.
Number of communications with valve closed	Once a day			
Number of registrations on the telecommunications network	1 per year			Includes where applicable the time required to obtain the IP channel. NOTE This excludes the very first connection of a SIM in a new installation.
Duration of remote data communication (with spontaneous)	80s	70s	60 s	Including any time needed to make the communication channel available.
Transmission power	P_{tx}=23 dBm			
Coverage Extension Level (ECL)	Level 2			
Local display operation	10 min / month			
Local interface operation	20 min / year			
Firmware update	3 in the useful life of the GdM			Communication times for FW update are not included in the above times.
Valve activation	30 cycles in the useful life			A cycle means an opening and a closing
Minimum battery life	**F**	**G**	**$Tstd_{batNBIOT}$**	

Tstd battery life values$_{batNBIOT}$, F and G must be declared by the manufacturer with reference to the parameters reported in table A.5.

The F and G values must be greater than $Tstd_{batNBIOT}$.

The value of $Tstd_{batNBIOT}$must be greater than $Tstd_{batGPRS}$

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APPENDIX B TIMES AND RULES FOR SYNCHRONIZATION FUNCTIONS

(regulations)

The technical specifications UNI/TS 11291-6:2013 and UNI/TS 11291-7:2013 define the minimum and maximum time deviations from the clock (before synchronization) and additional situations that must be taken into account to enable or prevent synchronization. They include additional rules that apply to the synchronization function.

Additionally, the above specifications define whether events should be logged or not, and additional options that apply to event queues, such as the alternative use of “synchronization logs”.

In case of synchronization, the following rules must also be applied:

1) All commands programmed in the past are executed immediately after receiving the command.

2) It must be ensured that no programmed event or command is lost or executed twice.

However, the application of these rules must not allow the extension of the validity of temporary keys or their revocation.

Note For example, when a forward synchronization exceeds the scheduled time for an event, that event must occur immediately. When a backward synchronization crosses the time of an event that has already occurred, that event must not be repeated.
This rule does not apply to pricing and consumption profile calculation functions, whose operation following synchronisation is described in UNI/TS 11291-6:2013.

APPENDIX C EXAMPLE OF BUILDING BUSINESS USE CASES

(information)

The following uses a use case for billing activities, based on the definition developed within the European mandate on smart metering. Figure C.1 shows the customer management service using the "Get scheduled meter readings" UC.

The instrument generally performs periodic data storage according to a schedule. These schedules are performed in the GdM before the instrument can start data acquisition. The schedule in question is defined by the customer management service.

The billing service must then retrieve the data acquired by the tool. This data is then linked to customer information and the billing plan to provide the invoice to the customer.

figure C.1 **Example of building business use cases**

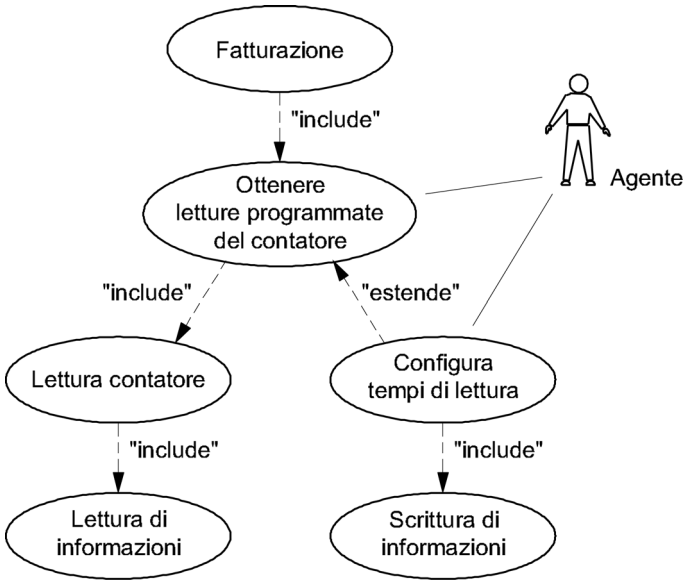
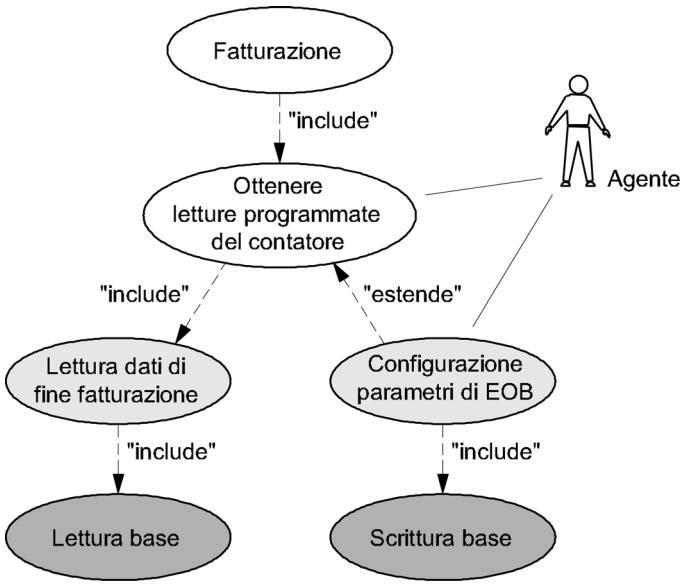


Figure C.2 shows how the measurement infrastructure management use cases described in this document relate to the use cases above. Use cases left blank are excluded from the scope of this technical specification.

figure C.2 **Example use cases related to measurement infrastructure management**



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APPENDIX D EKSF EXCHANGE FORMAT

(regulations)

Table D.1 defines the basic scheme for the EKSF exchange format.
Below is the schema of the XML file to be used (i.e. the corresponding XSD) and an example of a swap file.

table D.1 **EKSF Exchange Format**

Header	[[reference certificate]].pem /.ber /.pkcs12: certificate (base64 encoded)				Algorithm used			
	Certificate format		Certified content					
	Version of the Format	**Unique Identifier Device**	**Identifier Type Device**	**KEYC**	**KEYT**	**KEYN**	**Password**
	v0.1	YWZ-X-00-00-123456789	MU-G4 J	17C3CD32291A274E8D07037EA9616435	AAC55F0CAC5C51894D459573ED033E3A	F88A3139D440E632DCF391E1BB6E9460	ABCDEFGH
	v0.1	YWZ-X-00-00-123456788	MU-G4 J	17C3CD32291A274E8D07037EA9616435	4F5E30A7CD653340D0AAD521AF4654D0	8FC6AE0FC7BDBB2D506600B355CB152C	ABCDEFGH
	v0.1	YWZ-X-00-00-123456787	MU-G4 J	1BBD45D50DDEA2EA093F1747058C412A	082EB17D041758BE02BA3B2901A2F04C	CD5A69D366AD34E94473789B2912152A	ABCDEFGH
	v0.1	YWZ-X-00-00-123456786	MU-G4 J	3EDF85241F6FC29214F52C610C931AA0	ED2371BA7691B8DD4F0BD0932F6D7D25	A971F8FF54B8FC7F387B52FF21E39833	ABCDEFGH
	v0.1	YWZ-X-00-00-123456785	MU-G4 J	05CE65C202E732E101EF7740012947 8D	A073AF965039D7CB357BE5322017231E	09F4FCEE04FA7E770351A9A401FDCC2F	ABCDEFGH
	v0.1	YWZ-X-00-00-123456784	MU-G4 J	C17B3CFC60BD9E7E407E69A926B23F65	2CC181811660C0C00EEB2B2B08F3804D	F2189EBC790C4F5E50B2DF94306B52F2	ABCDEFGH
	v0.1	YWZ-X-00-00-123456783	MU-G4 J	22AA6DFA115536FD0B8E24A806EEE2CB	19027AB30C813D59085628E605007EF0	0701E21F0380F10F0255F60A0166C6D3	ABCDEFGH
	v0.1	YWZ-X-00-00-123456782	MU-G4 J	B9CE48D45CE7246A3DEF6D9C252941C4	15C136D90AE09B6C0740679D0459D7C5	90105C2348082E113005740B1CD0126D	ABCDEFGH
	v0.1	YWZ-X-00-00-123456781	MU-G4 J	15FFD2010AFFE900075546004665D33	E91AF128748D78944DB3A5B82E9EFD08	9BFF66BA4DFFB35D33FFCCE81F33148B	ABCDEFGH
Signature	5E005F00BC003300340003500D70009						

NOTE:
The format version is used to track (future) versions of this schema.
Unique device identifier: defined in UNI/TS 11291-12-2. Device type identifier: specific to each manufacturer.
Each key is individually encrypted with the recipient's public key; the entire file must be authenticated (signed) with the source's private key. The exchange format with the Guarantor Authority follows the same rules. The KEYC field is replaced by the KEYS field; the KEYT and KEYN fields are omitted.
The file format(s) is xml. In case the role of the Guarantor Authority is managed by the Distributor, the file format does not change, i.e. [KEYC, KEYT, KEYN] and [KEYS] must always be transferred with two different files according to the above format.

table D.2 **Description of the EKSF swap file in xsd format**

```
<xs:schema
        attributeFormDefault="unqualified" elementFormDefault="qualified"
        xmlns:xs="http://www.w3.org/2001/XMLSchema"
        xmlns:ds="http://www.w3.org/2000/09/xmldsig#"
        xmlns:xenc="http://www.w3.org/2001/04/xmlenc#">
    <xs:import namespace="http://www.w3.org/2000/09/xmldsig#" /    schemaLocation="http://www.w3.org
TR/2002/REC-xmldsig-core-20020212/xmldsig-core-schema.xsd"/>
    <xs:import namespace="http://www.w3.org/2001/04/xmlenc#" /    schemaLocation="http://www.w3.org
TR/xmlenc-core/xenc-schema.xsd"/>

<!--begin of document body (root tag) -->
<xs:element name="root">
    <xs:complexType>
        <xs:sequence>
        <!--Provides information about the content of the file. Optional -->
        <xs:element type="MetaInformationType" name="meta_information" minOccurs="0" maxOccurs="1" />

        <!--Content of the file. Mandatory -->
        <xs:element name="content">
            <xs:complexType>
                <xs:sequence>
                    <!--the content is only the list of security informations -->
                    <xs:element name="security_infos">
                        <xs:complexType>
                            <xs:sequence>
                                <!--
                                All the security information for a given device (see attribute
                                "unique_device_identifier")
                                Only one security information set per
                                Mandatory file (more than 1)
                                - - >
                                <xs:element    type="SecurityInformationType"    name="security_info"
minOccurs="1" maxOccurs="unbounded" />
                            </xs:sequence>
                            <!--The number of security information items shall be declared -->
                        <xs:attribute type="xs:unsignedInt" name="size"/>
                        </xs:complexType>
                    </xs:element>
                </xs:sequence>

                <!--The content is fixed to "keys" -->
                <xs:attribute type="ContentType" name="type" fixed="keys"/> </
                xs:complexType>
        </xs:element>
```

```
            <!--Signature of the file, as per XML dsig-->
            <xs:element name="Signature" type="ds:SignatureType" minOccurs="1" maxOccurs="1"/> </
            xs:sequence>
        </xs:complexType>
</xs:element>
<!--end of document body (root tag) -->
<!--
##############################################################################
# Hereafter the type definitions are exposed. No more document structure defined #
##############################################################################
- - >
<!--Representation of the information block that might be written on top of XML file -->
<xs:complexType name="MetaInformationType">
        <xs:sequence>
            <xs:element type="xs:string" name="version"/>
            <xs:element name="created_by">
            <xs:complexType>
                <xs:simpleContent>
                    <xs:extension base="xs:string">
                        <xs:attribute type="xs:string" name="organisation"/>
                        <xs:attribute type="xs:string" name="firstname"/>
                        <xs:attribute type="xs:string" name="lastname"/> </
                        xs:extension>
                </xs:simpleContent>
            </xs:complexType>
            </xs:element>
            <xs:element type="xs:dateTime" name="creation_date"/>
            <xs:element type="xs:dateTime" name="last_change_date"/>
            <xs:element type="xs:string" name="scope"/>
        </xs:sequence>
</xs:complexType>

<!--
        Representation of the logical device name for Italian market.
        Only one logical device name per device, thus it becomes the unique device identifier.
- - >
<xs:simpleType name="UniqueDeviceIdentifierType">
        <xs:restriction base="xs:string">
            <!--it must be 16 characters long -->
            <xs:minLength value="16"/>
            <!--allowed characters are ASCII letters and numbers only -->
            <xs:pattern value="([a-zA-Z0-9])+"/>
        </xs:restriction>
</xs:simpleType>

<!--
        The user name as defined in Blue Book 12.2, in class_id=15v2
- - >
```

```
<xs:simpleType name="UserNameType">
        <xs:restriction base="xs:string">
                <!--TODO: This pattern shall be modified to take into account the complete ASCII visible-string range -->
                <xs:pattern value="[a-zA-Z0-9]{1,4}"/>
        </xs:restriction>
</xs:simpleType>

<!--User login is comprised of an id (0-255) and a name -->
<xs:complexType name="UserLoginType">
        <xs:sequence>
                <xs:element name="id" type="xs:unsignedByte"/>
                <xs:element name="name" type="UserNameType"/>
        </xs:sequence>
</xs:complexType>

<!--Formal description of the password. Byte array with minimum length equal to 1 -->
<xs:simpleType name="PasswordType">
        <xs:restriction base="xs:hexBinary">
                <xs:minLength value="1"/> </
                xs:restriction>
</xs:simpleType>

<!--A Cosem password can be either in plaintext or encrypted -->
<xs:complexType name="CosemPasswordType">
        <xs:choice>
                <!--If encrypted, shall follow xml-enc definitions --> <xs:element
                name="encrypted" type="xenc:EncryptedKeyType" /> <!--If in plaintext,
                then shall fulfill requirements of "PasswordType" --> <xs:element
                name="plain_text" type="PasswordType" /> </xs:choice>

</xs:complexType>

<!--Format description of an AES128 key. Byte array with fixed length equal to 16 -->
<xs:simpleType name="Aes128KeyType">
        <xs:restriction base="xs:hexBinary">
                <xs:length value="16"/> </
                xs:restriction>
</xs:simpleType>

<!--A Cosem key can be eithr in plaintext or encrypted -->
<xs:complexType name="CosemKeyType">
        <xs:choice>
                <!-- If encrypted, shall follow xml-enc definitions --> <xs:element
                name="encrypted" type="xenc:EncryptedKeyType" /> <!-- If in plaintext,
                then shall fulfill requirements of "Aes128KeyType" --> <xs:element
                name="plain_text" type="Aes128KeyType" /> </xs:choice>

</xs:complexType>
```

```
<!--Choice to differentiate different contents of files -->
<xs:simpleType name="ContentType">
    <xs:restriction base="xs:string">
        <!-- if content tag has attribute type="keys", then content is targeted to key-exchange -->
        <xs:enumeration value="keys"/>
        <!-- if content tag has attribute type="command", then content is targeted to key-exchange -->
        <xs:enumeration value="command"/>
        <!-- if content tag has attribute type="response", then content is targeted to key-exchange -->
        <xs:enumeration value="response"/>
    </xs:restriction>
</xs:simpleType>

<!-- Possible types of devices within the application: Meter, Gateway and Repeater -->
<xs:simpleType name="DeviceType">
    <xs:restriction base="xs:string">
        <xs:enumeration value="GDM"/>                <!-- Meter -->
        <xs:enumeration value="DCGW"/>               <!-- Gateway -->
        <xs:enumeration value="REPEATER"/> </        <!-- Repeater -->
        xs:restriction>
</xs:simpleType>

<!--
    Specifies the basic content of the credentials for a single Cosem client
    - - >
<xs:complexType name="CosemClientCredentialType">
    <xs:sequence>

        <!-- only one Encryption key per client -->
        <xs:element type="CosemKeyType" name="encryption_key" minOccurs="0" maxOccurs="1" />

        <!-- only one Authentication key per client. Optional -->
        <xs:element type="CosemKeyType" name="authentication_key" minOccurs="0" maxOccurs="1" />

        <!-- Only one Password per client. Optional -->
        <xs:element type="CosemPasswordType" name="lls_password" minOccurs="0" maxOccurs="1" />

        <!-- User credentials. Needed only for Installer/Maintainer -->
        <xs:element name="user_logins" minOccurs="0" maxOccurs="1">
        <xs:complexType>
            <xs:sequence>
                <!-- If the user_logins is present, then at least one item must be present --> <xs:element
                type="UserLoginType" name="user" minOccurs="1" maxOccurs="10" /> </xs:sequence>

        </xs:complexType>
        </xs:element>
    </xs:sequence>
```

```
<!--
    Specifies the identifier of the Cosem client (ranges from 0 to 65535).
    This attribute must be present! -->
<xs:attribute type="xs:unsignedShort" name="id" use="required"/> </
xs:complexType>

<!--
    Description of a single set of security information for a given device. Name
    of device is reported in attribute "unique_device_identifier"

    A SecurityInformationType can be seen as a structure comprised of:
        - device type
        - device model
        - notes
        - master key
        - a list of CosemClientCredentials
    - - >
<xs:complexType name="SecurityInformationType">
    <xs:sequence>
            <!-- The device type is a well-defined string restricted to values  : GDM, DCGW and REPEATER -->
            <xs:element type="DeviceType" name="device_type"/>

            <!-- The device model is manufacturer specific. Optional -->
            <xs:element type="xs:string" name="device_model" minOccurs="0" maxOccurs="1" />

            <!-- The notes for each set of security info is manufacturer specific. Optional -->
            <xs:element type="xs:string" name="notes" minOccurs="0" maxOccurs="1" />

            <!-- This object specifies the Master key of the device. Optional -->
            <xs:element type="CosemKeyType" name="master_key" minOccurs="0" maxOccurs="1"/>

            <!-- This object holds all security credentials of a given Cosem client -->
            <xs:element type="CosemClientCredentialType"  name="cosim_client_credentials"
minOccurs="1" maxOccurs="unbounded" />
        </xs:sequence>
        <!-- Unique device identifier. Mandatory -->
        <xs:attribute  type="UniqueDeviceIdentifierType"  name="unique_device_identifier"
use="required"/>
    </xs:complexType>

    <!-- Type to code the message representing a protocol request -->
    <xs:complexType name="ProtocolRequestType">
        <xs:simpleContent>
            <!-- message content is expressed in ASCII hex -->
            <xs:extension base="xs:hexBinary">
            <!-- every message is numbered. It is up to creator of file to avoid duplicate numbers. -->
            <xs:attribute type="xs:unsignedInt" name="nr" use="required"/>
            </xs:extension>
        </xs:simpleContent>
    </xs:complexType>
```

```
<!-- Type to code the message representing a protocol response -->
<xs:complexType name="ProtocolResponseType">
    <xs:simpleContent>
        <xs:extension base="ProtocolRequestType">
        <!-- every message is provided with a reception timestamp. Timestamp is the one of the client -->
        <xs:attribute type="xs:dateTime" name="timestamp" use="required"/>
        </xs:extension>
    </xs:simpleContent>
</xs:complexType>

<!-- Type to code the rights of the Installer/Maintainer client -->
<xs:simpleType name="RightsType">
    <!-- It is a ASCII HEX binary string -->
    <xs:restriction base="xs:hexBinary">
        <!-- minimum length is 1, as it is defined in UNI/TS 11291-11-2 -->
        <xs:minLength value="1"/>
    </xs:restriction>
</xs:simpleType>

</xs:schema>
```

prospectus D.3 **Example of EKSF exchange file in xml format** <?

```
xml version="1.0" encoding="UTF-8"?>
<root xmlns:xenc="http://www.w3.org/2001/04/xmlenc#" xmlns:ds="http://www.w3.org/2000/09/xmldsig#">
    <meta_information>
        <version>1.2.3.4</version>
        <created_by organization="Organization"/>
        <creation_date>2014-12-24T13:32:44+01:00</creation_date>
        <last_change_date>2014-12-24T13:32:45+01:00</last_change_date>
        <scope>Exchange format of keys</scope>
    </meta_information>
    <content type="keys">
        <security_infos>
            <!-- First example, with encrypted keys -->
            <security_info unique_device_identifier="XXYY030013345678">
            <device_type>GDM</device_type>
            <master_key>
                <encrypted>
                    <xenc:EncryptionMethod Algorithm="http://www.w3.org/2001/04/xmlenc#rsa-1_5"/>
                    <ds:KeyInfo>
                        <ds:KeyName>Name of the security material employed to encrypt the field
xenc:CipherData</ds:KeyName>
                    </ds:KeyInfo>
                    <xenc:CipherData>
                        <xenc:CipherValue>VEVTVFRFU1RURVNUVEVTVA==</xenc:CipherValue> </
                        xenc:CipherData>
                </encrypted>
            </master_key>
            <!--Management -->
```

```
<cosam_client_credentials id="1">
<encryption_key>
    <encrypted>
        <xenc:EncryptionMethod Algorithm="http://www.w3.org/2001/04/xmlenc#rsa-1_5"/>
        <ds:KeyInfo>
            <ds:KeyName>Name of the security material employed to encrypt the field
xenc:CipherData</ds:KeyName>
        </ds:KeyInfo>
        <xenc:CipherData>
            <xenc:CipherValue>VEVTVFRFU1RURVNUVEVTVA==</xenc:CipherValue> </
            xenc:CipherData>
    </encrypted>
</encryption_key>
</cosem_client_credentials> <!--
Installer/Maintainer -->
<cosem_client_credentials id="3">
<encryption_key>
    <encrypted>
        <xenc:EncryptionMethod Algorithm="http://www.w3.org/2001/04/xmlenc#rsa-1_5"/>
        <ds:KeyInfo>
            <ds:KeyName>Name of the security material employed to encrypt the field
xenc:CipherData</ds:KeyName>
        </ds:KeyInfo>
        <xenc:CipherData>
            <xenc:CipherValue>VEVTVFRFU1RURVNUVEVTVA==</xenc:CipherValue> </
            xenc:CipherData>
    </encrypted>
</encryption_key>
<!-- optional field -->
<authentication_key>
<!-- In this example, the field "plain_text" is used. This allows to have unencrypted keys (eg
KEYT) -->
    <plain_text>000102030405060708090A0B0C0D0E0F</plain_text> </
    authentication_key>
<!-- optional field -->
<lls_password>
    <plain_text>ABCDEF01</plain_text> </
    lls_password>
<!-- optional field -->
<user_logins>
<!-- If field "user_logins" is employed, then at least one "user" shall be present. Up to a
maximum of 10 -->
    <user>
        <id>101</id>
        <name>XFGS</name>
    </user>
    <user>
        <id>102</id>
        <name>ABCD</name>
```

```
            </user>
            <user>
                <id>103</id>
                <name>A</name>
            </user>
        </user_logins>
        </cosem_client_credentials> <!--
        Guarantor Authority -->
        <cosem_client_credentials id="48">
            <encryption_key>
                <encrypted>
                    <xenc:EncryptionMethod Algorithm="http://www.w3.org/2001/04/xmlenc#rsa-1_5"/>
                    <ds:KeyInfo>
                        <ds:KeyName>Name of the security material employed to encrypt the field
xenc:CipherData</ds:KeyName>
                    </ds:KeyInfo>
                    <xenc:CipherData>
                        <xenc:CipherValue>VEVTVFRFU1RURVNUVEVTVA==</xenc:CipherValue> </
                        xenc:CipherData>
                </encrypted>
            </encryption_key>
        </cosem_client_credentials>
        <!-- Broadcasting -->
        <cosam_client_credentials id="32">
            <encryption_key>
                <encrypted>
                    <xenc:EncryptionMethod Algorithm="http://www.w3.org/2001/04/xmlenc#rsa-1_5"/>
                    <ds:KeyInfo>
                        <ds:KeyName>Name of the security material employed to encrypt the field
xenc:CipherData</ds:KeyName>
                    </ds:KeyInfo>
                    <xenc:CipherData>
                        <xenc:CipherValue>VEVTVFRFU1RURVNUVEVTVA==</xenc:CipherValue> </
                        xenc:CipherData>
                </encrypted>
            </encryption_key>
        </cosem_client_credentials>
    </security_info>
    <!-- Another example, filled with simplest possible data (keys are in plain_text, so complete file must
be encrypted) -->
    <security_info unique_device_identifier="XXYY030013345677">
        <device_type>DCGW</device_type>
        <!-- Management -->
        <cosam_client_credentials id="1">
            <encryption_key>
                <plain_text>000102030405060708090A0B0C0D0E0F</plain_text> </
                encryption_key>
        </cosem_client_credentials>
        <!-- Installer/Maintainer -->
```

```
                <cosam_client_credentials id="3">
                    <encryption_key>
                        <plain_text>000102030405060708090A0B0C0D0E0F</plain_text> </
                        encryption_key>
                </cosem_client_credentials> <!--
                Guarantor Authority -->
                <cosem_client_credentials id="48">
                    <encryption_key>
                        <plain_text>000102030405060708090A0B0C0D0E0F</plain_text> </
                        encryption_key>
                </cosem_client_credentials>
                <!-- Broadcasting -->
                <cosam_client_credentials id="32">
                    <encryption_key>
                        <plain_text>000102030405060708090A0B0C0D0E0F</plain_text> </
                        encryption_key>
                </cosem_client_credentials>
                <!-- Gateway -->
                <cosam_client_credentials id="64">
                    <encryption_key>
                        <plain_text>000102030405060708090A0B0C0D0E0F</plain_text> </
                        encryption_key>
                    </cosem_client_credentials>
            </security_info>
        </security_infos>
    </content>
    <!--File signature (may be present or not) -->
    <Signature>
        <ds:SignedInfo>
            <ds:CanonicalizationMethod Algorithm="http://www.w3.org/TR/2001/REC-xml-c14n-20010315"/>
            <ds:SignatureMethod Algorithm="http://www.w3.org/2000/09/xmldsig#rsa-sha1"/> <ds:Reference
            URI="">
                <ds:Transforms>
                    <ds:Transform Algorithm="http://www.w3.org/2000/09/xmldsig#enveloped-signature"/> </
                    ds:Transforms>
                <ds:DigestMethod Algorithm="http://www.w3.org/2000/09/xmldsig#sha1"/>
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twHNsyLPpW9FTeQ7Z3ftaXShKcyPeh6zOvMwDRKLxdQ=</ds:SignatureValue>
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                <ds:X509SubjectName> emailAddress=pippo@disney.com ,CN=PROPERTY OF THE KEY
NECESSARY FOR DECRYPTING THE CIPHERDATA,O=littlecryptographer,ST=RM,C=CL</ds:X509SubjectName>
                <ds:X509IssuerSerial>
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                <ds:X509IssuerName> emailAddress=pluto@disney.com ,CN=PlutoDisney,O=
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                    </ds:KeyValue>
                </ds:KeyInfo>
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        </root>
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